Assetto Maximus

A Curated Modlist & Guide for Assetto Corsa

Version 1.0 — July 2026

Compatible with Assetto Corsa v1.16.4 (August 2020)

Prepared for the driving enthusiast who wants a curated, stable, and progressively-enhancing modding experience — from stock to spectacular, one wave at a time.

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About This Guide

Assetto Maximus is a curated modlist and comprehensive game guide for **Assetto Corsa** (2014), version 1.16.4 (August 2020).

It is designed to take you from a completely vanilla installation to a fully-modded sim racing experience through progressive, carefully-tested waves.

Credits

- **Author**: The Assetto Maximus team
- **Mod Credits**: Each mod's respective author(s) — see individual mod pages for full credits
- **Special Thanks**: The Assetto Corsa modding community

Assetto Corsa is a product of Kunos Simulazioni. This guide is a community project and is not affiliated with or endorsed by Kunos Simulazioni.

Version

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Introduction

What is Assetto Maximus?

Assetto Maximus is a curated modlist and comprehensive game guide for **Assetto Corsa** (2014), version 1.16.4 (August 2020). It is designed to take you from a completely vanilla installation to a fully-modded sim racing experience through progressive, carefully-tested waves.

The project is built around three principles:

- **Stability first** — Every mod is verified for compatibility with AC v1.16.4. No conflicts, no broken builds.
- **Progressive complexity** — Each wave builds naturally on the previous one, introducing new mechanics and content only when you are ready for them.
- **No cheats** — No omniscient overlays, no physics exploits, no unfair advantages. Skill should be the limiting factor.

Wave Structure

Each wave pairs a curated modlist with a roleplaying narrative to give context to your progression:

Wave	Title	Theme
0	The Novice's Tale	Vanilla+ — learn the fundamentals
1	The Apprentice's Rise	Enhanced — deeper engagement

Future waves will be added as the guide evolves.

How to Use This Guide

1. **Start fresh** — Install Assetto Corsa from Steam (v1.16.4)
2. **Follow Wave 0** — Set up Content Manager, install UI/QoL mods, learn to drive
3. **Progress to later waves** — Add more mods as you gain experience
4. **Refer to appendices** — For troubleshooting, reference, and configuration details

Typographic Conventions

`Code / Command text` — Terminal commands, file paths, or keyboard shortcuts

Italic text — Emphasis or first mention of a term

[Blue underlined text]() — Clickable links

Installation Guide

Assetto Corsa — Base Installation

1. Purchase and download **Assetto Corsa** on Steam
2. Right-click the title in your library → **Properties** → **Betas** → select `1.16.4` from the dropdown
3. Wait for the download to complete
4. Launch the game once to generate configuration files, then exit

Verifying Your Version

In the main menu, check the bottom-right corner for version text. It should read `v1.16.4` (build date August 2020).

Content Manager

Content Manager (CM) is the essential launcher and mod manager for Assetto Corsa. It replaces the default launcher with a significantly more capable interface.

Download & Install

1. Visit the official CM site
2. Download the latest full release (not a patch)
3. Run the installer — it will detect your Assetto Corsa installation automatically
4. Launch Content Manager after installation

Initial Setup

1. Go to **Settings** → **Content Manager** → **Appearance**
   - Set **Theme** to your preference (Dark is recommended for readability)
   - Set **Language** if needed
2. Go to **Settings** → **Assetto Corsa**
   - Verify that the **Game path** points to your AC installation (e.g., `C:\Program Files (x86)\Steam\steamapps\common\assettocorsa`)
3. Go to **Settings** → **Customs**
   - **Cars search path**: Leave default
   - **Tracks search path**: Leave default

Installing Mods with CM

CM supports drag-and-drop installation:

- `.zip` / `.rar` / `.7z` archives containing car or track folders
- `.kn5` / `.kn` files (compiled Assetto Corsa data)
- Folders placed directly into `assettocorsa/content/cars` or `assettocorsa/content/tracks`

To install, simply drag the archive onto the CM window, or use **Install Mod** button.

Custom Shaders Patch (CSP)

Custom Shaders Patch is a major visual and functional enhancement for Assetto Corsa. It will be added in Wave 1.

Note for later: CSP versions must match AC v1.16.4. The last compatible pre-WeatherFX build is `0.1.79`. For later CSP versions with WeatherFX, use v1.16.4-post-launch compatible builds (`0.2.0`+).

Recommended Folder Structure

```
Steam/steamapps/common/assettocorsa/
├── content/
│   ├── cars/           — Car mods
│   ├── tracks/         — Track mods
│   └── fonts/          — Custom fonts (rare)
├── extensions/         — CSP configs
├── system/cfg/         — Game configuration files
└── assettocorsa.exe    — Game executable
```

Backup Your Installation

Before adding any mods, create a backup of your clean AC installation:

```
Copy "assettocorsa" folder to "assettocorsa-stock"
```

This will let you revert if anything goes wrong.

Wave 0: The Novice's Tale

"The checkered flag means nothing when you cannot find the racing line."

The Narrative

You have just passed your driving test. Your budget is modest — a compact hatchback, a second-hand set of tires, and a fire in your chest that matches the red of the setting sun. You sign up for your first track day with nothing but a helmet, a full tank, and the will to learn.

You do not need a faster car. You need to **become** faster.

Wave 0 Philosophy

Wave 0 is a **Vanilla+** experience. No game-changing mods. No physics overhauls. No content packs. The goal is to learn Assetto Corsa at its purest while improving the UI/UX to modern standards.

- **What changes:** Interface, HUD, quality of life
- **What stays stock:** Physics, content, graphics (as shipped)

Understanding the HUD

The default Assetto Corsa HUD displays:

- **Speedometer & Tachometer** — Upper corners
- **Gear indicator** — Center bottom
- **Lap timer** — Top center
- **Relative display** — Left side (shows nearby cars)
- **Rearview mirror** — Toggle with `F2`

Controls

Keyboard & Mouse (Basic)

Action	Default Key
Steer	`A / D`
Accelerate	`W`

Action	Default Key
Brake	`S`
Gear up / down	`Shift` / `Ctrl`
Look left / right	`Left Arrow` / `Right Arrow`
Look back	`Down Arrow`
Rearview mirror	`F2`
Toggle HUD	`Ctrl` + `H`
Change camera	`F1` (cycle)
Pause	`Esc`

Gamepad

Action	Xbox	PlayStation
Steer	Left stick	Left stick
Accelerate	`RT`	`R2`
Brake	`LT`	`L2`
Gear up / down	`A` / `X`	`×` / `□`
Clutch	`LB`	`L1`
Look left / right	Right stick	Right stick
Look back	Right stick click	`R3`
Rearview mirror	`Back`	`Share`
Toggle HUD	`Guide` + `H`	`PS` + `H`
Change camera	`Y`	`△`
Pause	`Menu`	`Options`

Tips for gamepad:
- Set **Steering Speed** to 50—70% in Content Manager controls for linear response
- Enable **Steering Filter** at low values (0—5) to smooth input
- Increase **Deadzone** to 5—10% if the stick drifts
- Use **Vibration** (enabled in Settings → Controls) for tyre slip feedback
- For throttle control: squeeze `RT` / `R2` gradually — abrupt application causes spin

Wheel Users

Configure your wheel in **Controls** → **Axis**. Key settings:

- **Steering lock**: Match your wheel's physical rotation (e.g., 900°)
- **Gamma**: 1.00 (linear response)
- **Filter**: 0 (no smoothing)
- **Force Feedback**: 100% gain (adjust per car)

See Configuration chapter for detailed FFB tuning.

Driving Curriculum: Phase 1 — Foundations

Goal: Stop crashing and build clean, repeatable laps.

1.1 Seating Position & Ergonomics

Before you turn a wheel, set up your physical environment. A correct seating position is the single cheapest performance gain available.

Wheel Position

- Sit so your shoulders remain relaxed against the seat back when your hands are at quarter-to-three
- Your elbows should be bent at 90—120°
- You should be able to turn the wheel full lock without leaning forward or stretching

Pedal Placement

- Your heel should rest on the floor, allowing your foot to pivot from the ankle
- The brake pedal should be reachable without lifting your heel
- For heel-toe downshifting: the throttle and brake should be close enough in height to blip with your right foot's edge

Field of View (FOV)

FOV is the single most important visual setting for accuracy. Too wide and distances compress (you brake too early). Too narrow and you lose peripheral awareness.

In ideal FOV: the dashboard width matches what you see in real life when sitting in the car.

To calculate your ideal FOV:

1. Measure your monitor width (in cm) and distance from your eyes to the screen
2. Use an online FOV calculator (set result as "Vertical FOV" in AC)
3. Or start with 35—45° (single monitor) and adjust until braking points feel natural

1.2 Vision Fundamentals

Your eyes lead the car. A common beginner mistake is staring at the front bumper or the track directly ahead.

Where to Look

- **Entry:** Look at the braking point, then the turn-in point
- **Mid-corner:** Look at the apex
- **Exit:** Look at the exit curb or track-out point
- **Between corners:** Look as far ahead as possible — ideally 2—3 turns ahead

Reference Points

Choose fixed objects (braking marker boards, curb colors, marshall posts) as reference points. Without reference points every lap is a guess; with them, you can measure and improve.

- **Brake marker:** "I brake when my front bumper passes the 100m board"
- **Turn-in:** "I turn when the curb pattern changes from red to white"

- **Apex:** “I clip the apex where the outer barrier has the yellow stripe”

1.3 Smooth Inputs

Jerky inputs upset the car’s balance. Every control movement should be gradual and deliberate.

Throttle

- Squeeze, do not stomp. Apply throttle progressively over about 0.3—0.5 seconds
- On exit, increase pressure as the steering straightens
- Abrupt lift-off mid-corner causes snap oversteer

Brake

- Apply pressure smoothly (not a stomp), building to maximum in about 0.1—0.2 seconds
- Release pressure gradually as you approach turn-in
- A sudden brake release unweights the front and reduces turn-in grip

Steering

- Avoid sawing at the wheel. Plan your steering input before you execute it
- Smooth, continuous rotations — not a series of small corrections
- If you need multiple corrections mid-corner, you turned in too early or too late

1.4 Understanding Grip Limits

Grip is finite. Every tyre has a maximum force it can deliver. Your job as a driver is to stay under that limit — and sense when you are near it.

The grip circle (or friction circle) concept:

- A tyre can deliver 100% grip total
- If you use 80% for braking and 80% for turning, you exceed 100% — the tyre slides
- **Traction pyramid:** Brake in a straight line, then turn, then accelerate. Never do two things at full effort simultaneously

Learn to feel the limit in practice: drive a circle in low gear and slowly increase speed until the car slides. This is the limit. Back off slightly — that is where speed lives.

1.5 The Racing Line

The fastest path through any corner follows the **outside-inside-outside** principle:

1. **Brake zone:** Stay wide on the outside — sets up a straighter braking zone
2. **Turn-in:** Aim for the apex (inside curb)
3. **Exit:** Let the car drift wide to the outside curb

Corner Types

- **90° turn:** Standard outside-inside-outside
- **Hairpin:** Very late apex to open up the exit
- **Sweeper:** Early apex, carry as much speed as possible
- **Chicane:** Connected left-right — sacrifice entry of the first to set up the second

1.6 Threshold Braking

Braking at the absolute limit of grip — the point just before the wheels lock — is called threshold braking.

Technique

1. Apply brake firmly but progressively
2. Listen to tyre squeal and feel pedal vibration (ABS off) or slight pedal pulsation (ABS on)
3. Reduce pressure slightly if you feel a wheel lock
4. The goal: maximum deceleration without locking

Without ABS

Most road cars in AC do not have ABS. Practice in the Mazda MX-5 Cup (ABS off):

- Apply brakes to about 80% initially
- Increase to near-lock as you feel the grip available
- If the front wheels lock, you cannot steer — release slightly to regain steering authority

1.7 Corner Phases

Every corner has three phases. Master each separately, then join them.

Entry (Brake Zone → Turn-in)

- Brake in a straight line
- Release brakes progressively
- Turn the wheel smoothly toward the apex

Mid-Corner (Apex)

- Steady throttle or coasting
- Minimum speed occurs here — feel for it
- Car should be settled, not sliding

Exit (Apex → Track-Out)

- Progressive throttle application
- Let the steering straighten naturally
- Track out to the outside curb

1.8 Recognizing Understeer vs Oversteer

Understeer (Push)

- **Feeling:** The front washes wide; you turn more but the car does not respond
- **Cause:** Too much speed at entry, or too much steering angle for the grip available
- **Fix:** Reduce entry speed, trail brake slightly, or add more rear brake bias

Oversteer (Loose)

- **Feeling:** The rear steps out; the car tries to spin

- **Cause:** Too much throttle mid-corner, abrupt lift-off, or too much entry trail braking
- **Fix:** Smooth throttle application, reduce trail braking depth, adjust rear wing or ARB

Learn to catch oversteer with steering correction (countersteer) — this will save you hundreds of times.

1.9 Building Consistent Lap Times

Clean laps are faster than spectacular laps. A spin costs 3—5 seconds. A four-wheel-off costs 1—2 seconds. A perfect lap with one mistake loses to a clean lap every time.

The Consistency Method

1. Run 10 laps without counting your time
2. Focus on hitting your reference points every lap
3. Once you can hit them 8/10 times, start timing
4. Your goal: 5 consecutive laps within 0.5 seconds of each other
5. **Then** push for outright speed

Practice Focus

- **Car:** Mazda MX-5 Cup or Abarth 500 (simple, forgiving)
- **Tracks:** Magione, Brands Hatch Indy, Monza Junior, Imola short layouts
- **Session:** 20—30 minute practice sessions — short enough to stay focused, long enough to build patterns

Installation Instructions

1. Download each mod from the link provided
2. Drag the `.zip` / `.rar` onto Content Manager
3. CM handles extraction and installation automatically
4. Enable each mod in CM’s **Mods** tab
5. Configure in-game overlays via CM’s **Settings** → **Assetto Corsa** → **Apps**

Verification

After installing Wave 0, launch AC through Content Manager. You should see:

- The Sidekick spotter reads out gaps and flags
- Car Radar shows a 360° proximity display
- Helicorsa highlights nearby cars with arrows
- Everything else is identical to stock Assetto Corsa

Wave 1: The Apprentice's Rise

"Speed is a byproduct of control. Learn control first."

The Narrative

Months of track days have paid off. Your lap times are consistent, your car control is instinctive, and you have caught the attention of a local club racing organizer. They offer you a seat in their amateur series.

Your hatchback is retired. A purpose-built track car sits in the garage. You have graduated from **driving** — now you learn to drive **fast**.

Wave 1 Philosophy

Wave 1 builds on the solid foundation of Wave 0. UI/QoL mods remain. Now we add:

- **Graphics enhancements** — CSP, weather, lighting
- **Content** — New cars and tracks
- **Practice infrastructure** — Telemetry tools, FFB optimization, data logging

Driving Curriculum: Phase 2 — Balance & Speed

Goal: Carry more speed by understanding weight transfer and car balance.

2.1 Weight Transfer Fundamentals

Everything in high-performance driving flows from weight transfer. Understanding where the weight is tells you where the grip is.

Under Braking

- Weight shifts **forward** — front tyres gain grip, rear tyres lose grip
- The car becomes more willing to turn (more front grip)
- The rear becomes less stable — lift-off oversteer risk is highest here

Under Acceleration

- Weight shifts **backward** — rear tyres gain grip, front tyres lose grip
- The car becomes less willing to turn (understeer on power)
- Traction control helps, but smooth throttle is better

In Transition

- Left-to-right weight transfer happens at turn-in
- The rate of transfer depends on how quickly you turn the wheel
- Aggressive steering = rapid transfer = potential spin

The **weight transfer pendulum**: Brake to load the front → turn in while releasing brake → the front has maximum grip exactly at turn-in. This is why releasing the brake and turning must overlap.

2.2 Trail Braking

Trail braking means carrying brake pressure **into** the corner, not releasing it all before turn-in. This keeps weight on the front tyres to improve rotation.

How to Practice Trail Braking

1. Brake in a straight line as normal
2. At the turn-in point, instead of fully releasing the brake, keep 10—20% pressure
3. As you turn the wheel, **gradually** release the remaining brake
4. The brake should reach zero at or just before the apex

Common Mistakes

- **Too much brake at turn-in** — This causes the rear to step out (spin)
- **Releasing too fast** — The car understeers because the front unweights suddenly
- **Trailing too deep** — Braking past the apex costs exit speed

2.3 Brake Release Timing & Modulation

How you release the brake is as important as when you apply it.

- **Linear release:** Steady, even pedal release — good for high-speed corners
- **Quick initial release, then slow:** Use when you need to stabilize the car before turning
- **Slow initial release, then quick:** Use when you want maximum front grip for a late turn-in

The brake release profile should match the corner. Experiment with different release shapes on the same corner and watch the lap time.

2.4 Throttle Application for Rotation & Traction

The throttle is not just for going faster — it is a car control tool.

Power-On Rotation

Applying throttle mid-corner shifts weight to the rear. In a rear-wheel-drive car, this can **tighten** the line (reduce understeer) by inducing slight oversteer at the rear. This is called **power oversteer** or **throttle steering**.

In a front-wheel-drive car, power-on causes **understeer** (the front washes wide). In this case, **lifting** the throttle shifts weight forward and tightens the line.

The Throttle Shapes

- **Squeeze:** Gradual increase from apex to exit — safe, predictable

- **Punch:** Early sharp application to rotate the car, then back off — advanced technique
- **Blip:** Quick throttle for downshift rev-matching (mandatory for non-sequential cars)

2.5 Slip Angle Awareness

The fastest way through a corner is not at zero slip angle (perfect grip) — it is at a small, controlled slip angle. The rear of the car should be working slightly to rotate.

The ideal slip angle is 2—6° of rear yaw. You can feel it as a gentle looseness — the car is "working" but not sliding.

To find it: drive a constant-radius corner in second gear. Add throttle until you feel the rear start to move. Reduce throttle slightly. That position — right at the edge of the slide — is where the car rotates fastest.

2.6 Managing Understeer & Oversteer Actively

In Phase 1 you learned to **recognize** understeer and oversteer. Now learn to **manage** them.

Correcting Understeer Mid-Corner

1. **Reduce steering angle** — Turning more makes it worse (front tyres are already at the limit)
2. **Slightly lift or hold throttle** — This transfers weight forward, loading the front tyres
3. **Or brake gently** — Trail-brake to force rotation

Correcting Oversteer Mid-Corner

1. **Countersteer immediately** — Point the front wheels where you want to go
2. **Lift or reduce throttle** — Treading carefully: complete lift can snap the car back the other way
3. **Catch and hold** — Maintain countersteer until the car stabilizes, then smoothly straighten

2.7 Minimum Corner Speed

The slowest point in any corner occurs at or just after the apex. Knowing where minimum speed occurs and what affects it is central to fast driving.

- **If the minimum speed is too low:** You are braking too deep, not trail-braking enough, or over-slowing
- **If the minimum speed is too high:** You will run wide on exit or lose traction on power
- **Target:** Minimum speed should occur at the apex, not before it

To check: use a telemetry trace and find the speed graph. The valley should align with the apex marker.

2.8 Using Kerbs Correctly

Kerbs are not obstacles — they are tools. But each type of kerb behaves differently.

- **Flat red/white kerbs:** Safe to drive on. Use them to extend the track width
- **Tall "sausage" kerbs:** Will upset the car. Avoid them except at very low speed
- **Painted kerbs (raised bumps):** Can unsettle the car. Two wheels on is fine; four wheels is risky

Kerb Technique

- **Entry kerb:** Two wheels on kerb to open up the corner
- **Apex kerb:** Clip the inside edge — do not let the car jump
- **Exit kerb:** Two wheels on to maximize track width on exit

2.9 Basic Telemetry Reading

Now that you have data logging tools (see Wave 1 modlist), you can compare laps.

The Three Traces

1. **Brake trace:** Should show a smooth, quick application followed by a controlled release. Any stuttering means inconsistent pedal work
2. **Throttle trace:** Should be smooth on application. An "on-off-on-off" pattern reveals nervous driving
3. **Steering trace:** Should show one smooth arc into the corner and one smooth arc out. Multiple corrections indicate poor entry planning

Comparing Laps

Overlay a reference lap (your best or a faster driver's) against a normal lap:

- Where is the brake trace different? (Braking earlier or later?)
- Where is the minimum speed different? (Carrying more or less speed?)
- Where is the throttle-on point different? (Getting on power sooner or later?)

Practice Focus: Phase 2

- **Cars:** Mazda MX-5 Cup, Toyota Supra MKIV, Porsche 911 GT3 RS
- **Tracks:** Imola, Brands Hatch (full), Suzuka — mid-tier technical tracks
- **Goal:** Trail-brake into every corner for a full session. You will spin — that is learning.

Driving Curriculum: Phase 3 — Cornering Precision

Goal: Extract maximum from every corner type.

3.1 Advanced Trail Braking & Rotation

Phase 2 introduced trail braking. Phase 3 refines it.

Late Trail Braking

On high-speed corners where you need maximum rotation, carry brake pressure deeper into the corner — past turn-in, toward the apex. This **forces** the car to rotate.

This is high-risk, high-reward. Too much and you spin. Too little and you understeer. The margin is narrow.

Rotation Points

Every car has a characteristic rotation point — the moment in the corner where the rear begins to rotate toward the apex. Identifying this point consciously lets you manipulate it:

- **Earlier rotation:** Trail brake deeper, turn in more sharply
- **Later rotation:** Release brake earlier, turn in more gradually

3.2 Spiral / Progressive Steering

Instead of turning the wheel to a fixed angle, use a **progressive** steering technique: increasing steering angle as you approach the apex, then decreasing on exit.

This creates a spiral path — the tightest radius at the apex, wider before and after.

How to Practice

1. On a long sweeper (e.g., Parabolica at Monza), enter with very little steering angle
2. **Gradually** add steering angle as the corner progresses
3. The maximum steering angle should coincide with the apex
4. On exit, gradually reduce steering

A constant steering angle through a corner means you are leaving time on the table — the car can turn tighter at the apex than at entry or exit.

3.3 Identifying Rotation Points

Your rotation point is where the car's yaw (rotation) matches the corner radius. Before this point, the car is still "straight." After this point, the car rotates toward the apex.

To find it: on corner entry, pay attention to when the car starts to face the apex. **That feeling** is your rotation point. Once you can feel it, you can control it with brake and throttle.

3.4 Double Apexes & Complex Corners

Some corners require two apexes because the road is not a simple arc.

Double Apex Technique

1. Clip the first apex on entry
2. Let the car drift slightly wide between apexes
3. Clip the second apex at mid-corner
4. Exit normally

Key insight: Do not try to carry a single line through a double-apex corner. Treat it as two connected corners — sacrifice a little between them to nail both apexes.

Examples: Suzuka's Spoon Curve, Nürburgring's Hatzenbach

3.5 Blind Corners & Commitment

Blind corners — where you cannot see the exit — require faith in your reference points. If you set your braking point and turn-in correctly, the car will be on the correct line even if you cannot see where it is going.

Rule: If you are not a little scared in a blind corner, you are not at the limit. Trust your marks, not your eyes.

Practice

Choose a blind corner (e.g., Eau Rouge at Spa, or the dip before the final corner at Bathurst). Cover the top half of your screen with tape. Drive the corner using only your reference points.

3.6 Elevation Changes & Camber Effects

Elevation

- **Uphill braking:** Less grip uphill because weight transfers rearward. Brake earlier
- **Downhill braking:** More grip downhill because weight loads the front. Brake later but more carefully — the car is unstable
- **Uphill corners:** Natural slowing effect — you can carry more speed
- **Downhill corners:** Natural speeding effect — you need to brake earlier

Camber

- **Negative camber (road tilts into corner):** More available grip. Carry more speed
- **Positive camber (road tilts away from corner):** Less grip. The car will want to run wide

Read the road: look at the surface angle on approach. If the road falls away, reduce speed. If it banks, trust it.

3.7 Compound Corners (Linking Multiple Turns)

Compound corners are sequences where the exit of one turn feeds directly into the entry of the next (e.g., the S's at Suzuka or Maggotts-Becketts-Chapel at Silverstone).

The Secret

You cannot take the optimal line through both corners. You must **compromise**:

- **First corner:** Take a line that sets up a good entry for the second
- **Second corner (or last in sequence):** Take the optimal line — this is where you get the time back

General rule: sacrifice entry of the first to maximize exit of the last.

3.8 Line Variation

In Phase 1 you learned the single racing line. Now learn to vary it.

Defensive Line

- Take the inside of the corner
- This sacrifices exit speed but protects the position
- Use when a car is alongside or behind

Attack Line

- Take the standard racing line (outside → apex → track-out)
- Maximizes exit speed
- Use when clear ahead or trying to pass on the next straight

Compromise Line

- A middle-ground line that preserves some exit speed while covering the inside
- Use in close racing where you do not know which side the attacker will take

3.9 Tyre Temperature & Pressure Management

At this level, you need to read the tyres — not just react to them.

Temperature Profile

- **Inner edge hotter than outer:** Too much negative camber — the car lacks straight-line braking grip
- **Outer edge hotter than inner:** Too little negative camber — the car lacks cornering grip
- **Center hotter than edges:** Over-inflated — reduced contact patch
- **Edges hotter than center:** Under-inflated — wallowing, poor response

Managing Wear

- Hot tyres lose grip. After 2—3 fast laps, the peak grip window closes
- In a hot lap: push on the first flying lap (peak grip)
- In a long race: manage pace to save tyres for the final stint

Practice Focus: Phase 3

- **Cars:** Porsche 911 GT3 RS, Ferrari 488 GTE, progressively faster machinery
- **Tracks:** Spa, Suzuka, Nürburgring GP, Laguna Seca — long circuits with diverse corner types
- **Goal:** Practice double-apex and compound corners specifically. Pick one complex sequence and run it 50 times

Basic Racing Terms

A common glossary of terms used throughout this guide and in the Assetto Corsa community.

Corner Phases

- **Braking zone** — The straight-line section before a corner where you reduce speed. Ends at the turn-in point. **Heard as:** “Brake at the 100m board.”
- **Turn-in** — The moment you begin steering toward the apex. Determines your entire corner trajectory. **Heard as:** “Turn in when the tree lines up with the kerb.”
- **Apex** — The point where your car is closest to the inside of the corner. Also known as the “clipping point.” Usually the slowest point of the corner. **Heard as:** “Clip the inside kerb at the second cone.”
- **Track-out** — The point where your car reaches the outside edge of the track on exit. Should maximize the usable width. **Heard as:** “Let the car run all the way to the outside kerb on exit.”
- **Exit** — The section from apex to track-out where you transition from steering back to straight-line acceleration. **Heard as:** “Get to full throttle before the exit kerb.”

Racing Lines

- **Racing line** — The optimal path through a corner that minimizes lap time. Typically: outside → apex → outside.
- **Geometric line** — The shorter path (straight line between entry and exit). Often slower because it sacrifices exit speed.
- **Late apex** — Moving the apex point further around the corner. Safer for passing and maintains more exit speed. Used when overtaking or in damp conditions.
- **Early apex** — Moving the apex point earlier. Allows a tighter entry but sacrifices exit speed. Used in slow, tight corners.
- **Compromise line** — A line that sacrifices optimal speed through one corner to set up a better line for the next corner in a sequence.

Car Balance Terms

- **Understeer (push)** — The front tyres lose grip before the rears. The car continues straight despite steering input. You turn more but the car does not respond.
- **Oversteer (loose)** — The rear tyres lose grip before the fronts. The car rotates more than intended. Requires countersteering to correct.
- **Neutral** — The ideal state where front and rear grip limits are balanced. The car rotates exactly as much as the steering input requests.
- **Weight transfer** — The movement of the car's mass under braking, acceleration, and cornering. Determines where grip is available.
- **Slip angle** — The angle between the direction a tyre is pointing and the direction it is actually moving. A small slip angle (2–6°) is optimal for cornering grip.
- **Trail braking** — Carrying brake pressure into the corner entry to keep weight on the front tyres and improve rotation.
- **Lift-off oversteer** — Sudden removal of throttle mid-corner causing weight to shift forward, unloading the rear tyres and inducing oversteer.

Race Terms

- **Delta time** — The difference in lap time between your current lap and a reference lap (your best, or the leader's). Shown as +/- time. Green = faster, red = slower.
- **Gap** — The time difference between your car and the car ahead or behind. Gap to ahead shrinks when you are faster. Gap to behind grows when you are slower.
- **Blue flag** — A flag shown to slower cars indicating a faster car is approaching to lap them. The slower car must hold their line and allow the pass.
- **Track limits** — The defined edges of the track. Exceeding them (all four wheels beyond the white line or kerb) may result in a warning or time penalty depending on the server or championship rules.
- **Pit window** — The optimal lap range for making a pit stop. Entering too early or too late can cost track position.

Setup Terms

- **Camber** — The vertical angle of the tyre relative to the road surface. Negative camber (top of tyre tilted inward) improves cornering grip.
- **Caster** — The steering axis angle. More caster improves straight-line stability and steering return.
- **Toe** — The angle of the tyres relative to the car's centreline. Toe-in (fronts pointing inward) improves stability. Toe-out improves turn-in response.
- **ARB (Anti-Roll Bar)** — A suspension component that resists body roll. Softer = more mechanical grip, stiffer = quicker response.

- **Ride height** — The distance between the chassis and the ground. Lower = better aerodynamics, stiffer suspension feel. Higher = more suspension travel, better kerb riding.
- **Dampers (shocks)** — Control the speed of suspension compression (bump) and extension (rebound). Affect how the car responds to kerbs, elevation changes, and weight transfer.

The Perfect Lap

A perfect lap is not one magic corner — it is connecting every phase of every corner correctly, lap after lap. This section breaks down the ideal sequence and explains how to build it visually, corner by corner.

The Four Phases of a Corner

Every corner has four distinct phases. Time lost in any phase cannot be fully recovered in the next.

	Phase 1	**Phase 2**	**Phase 3**	**Phase 4**
Name	Brake	Turn-in	Apex	Track-out
Action	Maximum brake	Trail brake + steering	Min speed + feed throttle	Full throttle + unwind
Weight	Forward	Front loaded	Neutral	Rear loaded
Speed	Decreasing	Low point	Minimum	Increasing
Risk	Lock-up	Spin	Understeer	Oversteer

Table 1: The four phases of a corner. Each phase has a distinct objective, weight state, speed profile, and risk. Mastering the phase transitions is how you build a perfect lap.

1. **Braking zone** — The car should be in a straight line. Brake at maximum pressure (threshold braking) then begin to release smoothly. The brake release transitions into…
2. **Turn-in** — Steering input begins while the brake is still 10–20% applied (trail braking). Weight is on the front tyres, giving maximum rotation. The steering input should be progressive, not jerky.
3. **Apex** — Brake reaches zero at or just before the apex. The car is at its minimum speed and tightest radius. Throttle application begins here — gently at first, then aggressively.
4. **Track-out** — The car reaches the outside edge of the track. Steering is fully unwound. Throttle is at 100% if the car can take it. The earlier you reach full throttle, the higher your exit speed and the faster the following straight.

The Corner Speed Tradeoff

The fundamental compromise in cornering:

- **Entry speed vs Exit speed** — Every km/h you carry into the corner costs you more on exit. The car needs more steering angle and more rotation to handle higher entry speed, which delays throttle application. Typically, sacrificing entry speed for earlier throttle is the faster strategy.
- **Minimum speed** — The slowest point should be at the apex, not before it and not after it. If your minimum speed occurs before the apex, you braked too early or turned in too late. If it occurs after the apex, you turned in too early or applied throttle too aggressively.
- **The principle:** Slow in, fast out. Every corner you enter slower than you think you need to will reward you with a higher exit speed. Trust that the extra exit speed will more than compensate for the entry speed you gave up.

	Fast In (Red)	Slow In (Green)
Entry	High speed ~ over-slowed	Slightly lower speed
Apex	Low (deep valley)	Higher (clean apex)
Exit	Low (late to full power)	High (early to power)
Result	Slower overall	Faster overall

Table 2: The speed tradeoff. **Fast In (Red):** high entry speed forces late braking and a low apex speed — the exit is compromised. **Slow In (Green):** sacrificing entry speed maintains a higher apex and reaches full throttle earlier. Green is almost always faster overall.

Building a Perfect Lap Step by Step

1. **Start with a reference** — Pick a track and car you know well. Do 5 laps at 80% effort. Your goal is consistency, not speed.
2. **Improve one corner at a time** — Identify the corner where you lose the most time (use telemetry or to feel). Spend 10 laps working on only that corner. Entry speed? Brake point? Throttle application point?
3. **String corners together** — Once each corner is individually correct, focus on transitions between corners. Can you carry the exit speed from Turn 3 into the braking zone of Turn 4? If not, your line through Turn 3 is compromising Turn 4.
4. **The confidence lap** — A perfect lap requires trust. Trust that the car will grip. Trust that the apex will be there. Trust that you can open the throttle earlier. This confidence only comes from practice — there is no shortcut.
5. **Chase the feeling, not the number** — When you complete a lap that feels effortless — smooth inputs, no corrections, no fight — check the lap time. It will often be your fastest. That feeling is the target state. Reproduce it.

Finding Your Limit

- If you never spin or go off-track, you are driving below your limit.
- If you spin or go off-track every session, you are exceeding your limit.
- The target: one spin per practice session. That means you touched the limit and found it. Next session, you know where it is.

Tuning Reference

Tyre Pressures

- **Target range:** 26—28 PSI when hot
- **Too low:** Car feels sluggish, tyres overheat quickly
- **Too high:** Car feels twitchy, reduced contact patch
- Check pressures after 3—4 laps of consistent driving

Camber

- Negative camber improves cornering grip at the expense of straight-line braking
- **Start with:** −2.5° front, −1.5° rear
- **Adjust:** If the car understeers at corner entry, add more front negative camber

Anti-Roll Bars (ARB)

- Softer bar = more mechanical grip, more body roll
- Stiffer bar = less grip transfer, quicker response
- **Understeer fix:** Soften front ARB or stiffen rear ARB
- **Oversteer fix:** Stiffen front ARB or soften rear ARB

Installation Sequence

1. Ensure Wave 0 is fully installed and verified
2. Install **Custom Shaders Patch** via Content Manager
3. Install **Pure** (weather controller — the successor to Sol; requires CSP)
4. Install all other Wave 1 mods in any order
5. Verify CSP is active

Wave 2: The Master's Ascent

"When you are no longer fighting the car, you can fight the race."

The Narrative

You have the speed. You have the consistency. Now you need to **win**.

A professional coaching program has taken notice of your results. You are invited to an elite driver development camp — a week of advanced techniques, race simulations, and mental conditioning. The other drivers are just as fast as you. The margin between victory and defeat will not come from raw pace.

It will come from craft.

Wave 2 Philosophy

Wave 2 completes the Assetto Maximus Driving Curriculum. All previous mods remain. New additions focus on **racing infrastructure**:

- **Advanced telemetry & data analysis** — Full MoTeC-style lap comparison
- **Weather & dynamic conditions** — Pure weather scripts
- **High-fidelity content** — Premium car and track mods (if desired)
- **AI & race practice tools** — AI enhancement, grid starts

Driving Curriculum: Phase 4 — Race Craft & Mastery

Goal: Race intelligently and drive at the limit consistently.

4.1 Overtaking Techniques

The Setup

Overtakes are won before the braking zone. The actual pass is just the execution.

1. **Close the gap** — Use slipstream on the preceding straight to pull alongside
2. **Choose your side** — Commit to inside or outside before the braking zone
3. **Claim the corner** — You must be at least door-to-door before the apex

The Slipstream Pass

Follow within 0.3—0.5 seconds on a straight. The drag reduction gives you 5—10 km/h extra at corner entry. Brake slightly later and roll the inside line.

The Switchback

Fake an inside move to force the defender to cover the inside. At the last moment, change to the outside line. The defender, having committed to the inside, will have a slower exit — you carry more speed on the outside.

The Late Brake

Brake later than the defender, but **control** that brake — a dive-bomb that misses the apex and bumps the other car is not a pass; it is a crash. You must make the corner on the racing line, even if you are side-by-side.

4.2 Defensive Driving

The Single Move Rule

You may make one defensive move per braking zone. Once you choose your line, you must hold it. Weaving or moving twice is illegal and dangerous.

Inside Defense

Cover the inside line before the braking zone. This forces the attacker to go around the outside. You sacrifice exit speed (slower line through the corner) but protect the position.

Outside Defense

Stay on the racing line — do not cover the inside. This invites the attacker to try the inside, where they have to brake earlier. On exit, you have more speed and can pull away.

When to Yield

If an attacker has significant overlap before the apex, you must leave them space. Fighting a lost position often loses two positions — yield cleanly, fall in behind, and look for a counter-attack on the next straight.

4.3 Traffic Management & Awareness

Approaching Traffic

When catching slower cars (lapping):

1. Do not assume they see you — use your spotter (Sidekick / Crew Chief)
2. Choose a safe passing zone — ideally a straight with a long braking zone
3. **Point-by**: At the passing zone, point to the side you want them to let you through
4. Make the pass clean and immediate — do not race them

Being Lapped

1. Hold your line — do not weave or brake unpredictably
2. Lift slightly on the straight to let the faster car by

3. Stay on the racing line and let them choose the passing side
4. Once passed, fall in behind — do not try to re-pass

Multi-Class Awareness

In mixed-class racing (e.g., GT3 + GTE + slower classes):

- Faster classes are responsible for safe overtaking
- Slower classes should hold their line and be predictable
- Use your radar (Car Radar) and spotter to track closing speed differentials

4.4 Wet Weather Driving

The Wet Line

In the dry, you want the racing line (maximum rubber = maximum grip). In the wet, the racing line is **slippery** — rubber becomes grease when wet.

- **Avoid** painted lines, rubber buildup, and polished asphalt
- **Aim for** concrete patches, curb edges (where water drains), and less-traveled lines
- Braking zones shift: brake on areas with better drainage

Technique Adjustments

- **Brake earlier and smoother** — Threshold braking is dangerous in the wet
- **Feather the throttle** — Power oversteer happens at much lower speeds
- **Look further ahead** — Grip changes are easier to spot if you scan 5—10 seconds ahead
- **Increase following distance** — Spray reduces visibility to zero

Wet Line Variation

In a heavy rain race, the optimal line changes every lap as the track evolves. The driver who adapts fastest wins. Practice this:

1. Start a race on a dry track with Pure weather progression (time accelerated)
2. As rain begins, consciously search for grip on alternative lines
3. Note where standing water forms on the track — these become permanent danger zones

4.5 Drivetrain Characteristics

Each drivetrain behaves differently at the limit. Adapt your technique accordingly.

Front-Wheel Drive (FF)

- **Understeer is the default** — Throttle on corner exit pushes the nose wide
- **Lift to rotate** — Lifting the throttle shifts weight forward, helping turn-in
- **No power oversteer** — The front tyres do both the turning and the driving
- **Trail brake aggressively** — The rear is light; trail braking helps rotation

Rear-Wheel Drive (FR / MR)

- **Oversteer is available** — Power oversteer on exit, lift-off oversteer on entry
- **Throttle steering** — Use the throttle to tighten or widen your line mid-corner
- **Trail brake carefully** — Too much trail brake and the rear steps out

- **Smooth throttle essential** — Abrupt application = spin

All-Wheel Drive (AWD)

- **High entry grip** — AWD understeers less on entry than RWD
- **Power-on neutrality** — Add throttle mid-corner without spinning the rears
- **Understeer on exit** — The front tyres push when you apply power early
- **Brake later** — AWD allows later braking due to engine braking on all four wheels

Mid-Engine Rear-Wheel Drive (MR)

- **Extreme rotation** — The engine mass is behind the driver; turn-in is very sharp
- **Snap oversteer** — If the rear breaks loose, it happens fast — countersteer immediately
- **Throttle-sensitive** — Small changes in throttle position have large effects on balance
- **Smooth, smooth, smooth** — MR cars punish aggressive inputs

4.6 Class-Specific Techniques

GT3 / GTE

- ABS and traction control are available but should be set to minimum intervention
- Use trail braking heavily — these cars rotate well under brakes
- Kerb usage is aggressive — flat curbs are part of the racing line

Formula / Open-Wheel

- No ABS, no TC — pure driver input
- Braking is later and harder — high downforce means huge grip under brakes
- Steering input must be minimal — too much steering angle creates drag
- Downforce dependency: these cars have less grip at low speed, more at high speed

Road Cars / Touring Cars

- Minimal aero — mechanical grip is everything
- Slower corner entry, earlier power application
- Body roll is your feedback — feel it through the seat, not the wheel
- Mismatched tyre temps are a bigger issue than in downforce cars

4.7 Brake Bias Adjustment While Driving

Brake bias controls how much braking force goes to the front vs rear wheels. Adjusting it mid-race is an advanced skill that lets you adapt to changing conditions.

When to Change Bias

- **More rear bias (move bias backward):** When front tyres are overheating, or when the car understeers at corner entry
- **More front bias (move bias forward):** When the car is unstable under braking, or at high-speed tracks with long braking zones
- **Rain / low grip:** Move bias rearward to prevent front lock-ups
- **Low fuel (end of race):** Light car = less front grip. Move bias rearward

How Much

Adjust in 1—2% increments. A 5% change can transform the car's behavior.

4.8 Active vs Passive Driving Style

Passive Driving

Letting the car do the work. Minimal steering correction, predictable inputs, smooth transitions. This is fast when the car is well-balanced and the tyres are fresh.

Active Driving

Constantly managing the car's balance with brake, throttle, and steering corrections. The driver **creates** rotation, manages slip angle, and adjusts the line to track conditions. This is necessary when:

- The car is not perfectly balanced (always)
- Tyres are degrading
- Track conditions are changing (temperature, grip levels, rain line)
- Fighting for position

Great drivers are active, not passive. They are always working.

Mental State: Flow vs Overdrive

- **Flow:** The car and driver are in sync. Decisions happen without conscious thought. Laps feel effortless. This is the target state
- **Overdrive:** Trying too hard. Inputs become jerky. The driver is fighting the car. Lap times get worse, not better

If you find yourself overdriving: slow down. Do 3 laps at 80% effort. Reset your rhythm. Then build back up.

4.9 Mental Game

Focus Management

- A race is 30—60 minutes of sustained concentration. Your brain will fatigue.
- **Before the race:** Visualize the first 3 corners and the first lap
- **During the race:** Break the race into segments (quarters, or chunks of 5 laps)
- **Mistake recovery:** The moment after a mistake is the most dangerous — you are tempted to overcompensate. Reset: take the next corner at 90% to regain composure

Pressure Handling

- If a driver is pressuring you from behind: focus on the corner ahead, not the mirror
- If you are pressuring a driver ahead: be patient. The pass does not need to happen in the next corner. The race is long
- **Countdown method:** "I will attack for 5 corners. If I do not pass, I will drop back, cool the tyres, and try again in 2 laps"

Recovery from Mistakes

A spin, a lock-up, or an off-track moment does not end the race — unless you let it affect the next 10 corners.

1. Acknowledge the mistake
2. Tell yourself: “That lap is gone. The next one starts now.”
3. Do one gentle corner to reset your rhythm
4. Build back to full attack over 2—3 corners

4.10 Full Telemetry Analysis & Self-Coaching

You have the tools (see Wave 2 modlist). Now use them to coach yourself.

Session Workflow

1. **Drive 5—10 clean laps** — Do not look at the timer
2. **Identify your best lap** — By feel, then confirm with lap time
3. **Overlay all laps** — Find the corners where your best lap gained time
4. **Compare to a reference** — Use a faster driver’s lap (or your own target lap)
5. **Focus on one corner** — Do not try to improve every corner at once

What to Look For

- **Speed trace:** Where is my minimum speed lower than the reference?
- **Brake trace:** Am I braking earlier or later? Am I releasing smoothly?
- **Gear trace:** Am I in the right gear through the corners? Is my shift point optimal?
- **Steering trace:** Is there a “flat spot” where I hold constant steering (understeer)?

Self-Coaching Questions

- “Where am I losing the most time?”
- “Is this a confidence issue (I could carry more speed but I am afraid) or a skill issue (I do not know how to carry more speed here)?”
- “If I could change one thing about my driving today, what would it be?”

Tool-by-Tool Guide

These tools were installed across Waves 0–2. Here is how to use each one to find time, solve problems, and build racecraft. Open them one at a time during practice sessions until reading them becomes automatic.

ACTI is the bridge between AC and MoTeC i2 Pro, the professional data analysis software used by real race teams. This is the deepest analysis tool available.

What it records: Speed, throttle position (%), brake pressure, steering angle, gear, RPM, individual wheel speeds, suspension travel, damper velocity, tyre temps, and G-forces — at up to 100 Hz.

How to enable: In Content Manager, go to **Settings** → **Custom Shaders Patch** → **General Patch Settings** → **Enable telemetry logging**. ACTI also needs its `.exe` running in the background during sessions.

Workflow:

1. Drive 5–10 consistent laps. Do not push to the limit – leave 2% for clean data. Pushing 100% introduces noise (slides, corrections) that obscures the signal.
2. Open MoTeC i2 Pro. File → Open → navigate to `Documents/Assetto Corsa/aim/telemetry/` and load your `.ld` file.
3. **First pass — speed trace overlay:** Open a Speed vs. Distance graph. Overlay your 3 best laps (Ctrl+click multiple laps in the session browser). Where the lines diverge is where you are inconsistent or have opportunity.
4. **Second pass — brake trace:** Look at the Brake graph for your best lap. Is the trace a sharp triangle (hard initial hit, smooth linear release) or a trapezoid (too much brake too long, late release)?
5. **Third pass — corner-by-corner:** Pick the corner you lose most time on. Look at:
   - **Minimum speed:** If your minimum speed is 5 km/h lower than your other laps, you are braking too early or releasing too late.
   - **Throttle application point:** The moment you get back to 100% throttle. Every tenth of a second later than necessary costs 3–5 metres down the following straight.
   - **Steering trace flat spots:** If the trace plateaus through a corner, the front tyres are understeering. Either the entry speed is too high or you need a setup change.
6. **Fourth pass — compare to reference:** If you have a faster driver's telemetry file, overlay it. Find exactly where their brake point differs from yours. Try moving your brake point by 5 metres in the same direction.

Finding time:
- Brake trace too long → you are trailing too deep. Release faster.
- Throttle trace takes 0.3s to go from 0→100% → you are squeezing the throttle. Try a more aggressive initial squeeze; trust the rear tyres.
- Minimum speed 3 km/h higher but lap time the same → you are over-slowing the exit. Trade entry for exit.
- Steering trace oscillating through a corner → you are making micro-corrections. Smooth out the arc.

Pro tip: Do not open MoTeC before you have driven. Analyse AFTER the session. Telemetry viewed during a session creates "analysis paralysis" – you stop driving by feel.

Sidekick is an in-game app that shows live tyre and brake temperatures per wheel, plus delta comparison to your best lap and multi-lap tyre history.

What to watch:

- **Tyre temps:** Each compound has an optimal temperature window (typically 80–100 °C for slicks, 70–85 °C for semi-slicks). Green = in the window. Blue = cold (no grip, understeer, locking fronts). Red = overheating (greasy, sliding, graining). Load the app and glance at it during straights.
- **Pressure:** Building? Tyres warming up. Dropping? Sudden pressure loss = puncture.
- **Delta display:** The +0.00 / −0.00 number compares your current sector time to your best sector. Red positive = slower than best. Green negative = faster. Focus on the delta through ONE corner at a time.

How to use it to go faster:

1. **Entry phase:** Watch tyre temps on the straight before braking. If fronts are blue and rears are green, move brake bias forward 1–2% to put more heat into the fronts.
2. **Mid-corner:** Check the delta. If you exit a corner and the delta turns green (negative), you carried more speed **through** that corner than before. Note what you did differently.
3. **Stint management:** After 10 laps, check if any tyre is running 10+ °C hotter than the others. That tyre is overworking – either you are asking too much of that corner of the car, or the setup has a bias problem. On the same lap, check if the delta is rising across the stint – you may be overheating the compound and losing pace.
4. **Brake temps:** Sidekick also shows brake disc and pad temps. Front brakes should be 100–200 °C hotter than rears. If rears match fronts, your brake bias is too far rearward.

Finding time with Sidekick:

- Front tyres blue, rear tyres green after 3 laps → brake bias too far rearward or you are not braking hard enough to load the fronts.
- Inside tyre across a corner is 20 °C colder than the outside tyre → camber is wrong or you are not loading that tyre through the corner (driving too conservatively).
- Delta goes green in corners with heavy braking but red in fast sweepers → your strength is braking zones, your weakness is cornering speed. Spend your next practice session on the fast sweepers.

Camber Extravaganza shows real-time camber angle for each wheel, updated continuously through corners. This is the only tool that tells you whether your static camber settings are working dynamically.

What it shows: Four numbers (FL / FR / RL / RR) each showing current camber angle in degrees. Negative values = top of tyre leans inward (this is what you want for cornering). Values update in real time.

How to read it:

1. Drive a lap at 90% pace on a track with varied corners.
2. **Straight line camber:** Glance at the values on a straight. This is your static camber. For a GT3 car, you want approximately −3.0° to −4.0° front, −2.0° to −3.0° rear.
3. **Mid-corner camber:** Glance at the values at the apex of a long corner. The outside tyres should gain approximately 1–2° more negative camber (e.g., from −3.5° to −5.0°). This is the suspension compressing and the body rolling, which adds dynamic camber.
4. **The goal:** The outside tyre through a corner should have approximately −3.0° to −4.5° of camber relative to the track surface. The tyre contact patch should be flat on the road.

Signs of incorrect camber:

- **Outside tyre shows −1.5° or less mid-corner:** You have too little static camber. The tyre is rolling onto its outside shoulder. This causes understeer and wears the outer tread. Add −0.5° front camber and test again.
- **Outside tyre shows −6.0° or more mid-corner:** You have too much static camber. The tyre is riding on its inside edge. You lose contact patch area and have less grip. Reduce camber by −0.5°.
- **Inside tyre shows positive camber through a corner:** The inside wheel is lifting or the anti-roll bar is too stiff – it is pulling the inside wheel off the road surface.
- **Camber change from straight to corner is less than 0.5°:** The car is not rolling. Either the springs/ARBs are too stiff, or you are not carrying enough speed through that corner to generate cornering force.

How to use it to improve setup:

1. Do 3 consistent laps. Record mid-corner camber values at 3 key corners (one slow, one medium, one fast).

2. If camber is off in slow corners but good in fast corners, the issue is static camber, not dynamic. Adjust camber settings.
3. If camber is good in slow corners but off in fast corners, the issue is body roll. The car rolls too much at high speed. Stiffen the front ARB by one step.

Finding time with Camber Extravaganza: Every 0.5° of camber error costs approximately 0.2–0.4 seconds per lap on a medium-length circuit. Getting camber right is one of the highest-return setup changes you can make.

[Car Radar]() is the advanced successor to Helicorsa. It shows cars around you in a 360° ring with directional vectors and closing speed indicators.

How to configure:

- **Detection range:** Set to 15 metres for single-class racing, 50 metres for multi-class (LMPs closing on GTs at 350 km/h need early warning).
- **Blind spot warning:** Enable. Turns the corresponding arc red when a car is alongside your rear quarter panel.
- **Opacity:** Set to 30–40% – visible but not distracting. Position it near the centre of the screen where your peripheral vision catches movement.

How to read it:

- **Solid dot ahead:** Car directly in front. Dot grows = you are closing. Dot shrinks = they are pulling away.
- **Arrow moving clockwise/counterclockwise:** Car moving alongside you (left or right side). The arrow's speed shows their relative motion.
- **Flashing red arc:** Blind spot occupied. Do not move in that direction.
- **Rapid dot approaching from behind:** Faster class car closing on you. Hold your line and let them choose the pass.

How to use it in races:

1. **Entry to braking zone:** Before braking, glance at radar. If a dot is alongside on the inside, leave a car's width – they are committing to the inside line.
2. **Exit of corner:** As you accelerate, check radar for cars that went wide and are re-joining. Give them space.
3. **Multi-class traffic:** When you see a rapidly closing dot from behind, check which side they are approaching from. If they are on your right, move slightly left (predictably, not abruptly) to give them the preferred inside line.
4. **Start of race:** Car Radar is most valuable at race starts. Set the detection range to 10 metres for T1 chaos. The radar shows cars on both sides that you cannot see in mirrors.

Finding time with Car Radar: This is a safety tool, not a speed tool. But clean racing IS faster racing. Every unnecessary collision costs 5–30 seconds. Avoiding one crash per race is worth more than any setup change.

[Race Essentials]() is a comprehensive HUD that replaces the need for multiple individual apps. Think of it as your race engineer on screen.

What it shows: Position, gap to car ahead/behind, fuel remaining (laps), last lap time, best lap time, stint length, tyre life estimate, pit stop countdown.

How to use it for race management:

- **Fuel calculator:** Before a race, the app estimates laps of fuel remaining based on your current consumption rate. If it says "6 laps" and you have 8 laps to go, lift and coast on straights (0% throttle, in gear) to save fuel. Lifting for 100 metres on a straight saves 0.05–0.1 laps of fuel.

- **Gap to ahead/behind:** Do not chase the car ahead if the gap is over 3 seconds and shrinking by less than 0.2s/lap. You will overheat your tyres and lose pace to the car behind. Maintain pace instead.
- **Gap to behind:** If the car behind is gaining 0.5s/lap and there are 5 laps left, they will catch you. Plan your defense: which corners will you protect? Do you have a straight-line speed advantage? Use the final 2 laps to save tyres for the last-lap battle.
- **Pit stop timer:** When pitting, Race Essentials shows time lost in pit lane vs. competitors who stay out. This tells you whether the undercut or overcut is working.

Finding time with Race Essentials:

- Fuel weight costs approximately 0.3 seconds per lap per 10 kg. A 15-lap race starting with 60L of fuel means you are 1.8 seconds slower on lap 1 than you will be on lap 15. Your target lap time should drop across the stint. If it does not, you are not exploiting the lightening car — push harder on the final 5 laps.
- The gap to the car ahead tells you whether you are faster. If the gap is shrinking by 0.3s/lap, you have a pace advantage. Wait for them to make a mistake — they feel the pressure too.

Both are installed. Here is the distinction:

- **Helicorsa:** Simpler, cleaner, lower cognitive load. Best for single-class sprint racing (GT3, F3, MX-5 Cup) where closing speeds are similar and you need quick left/right awareness.
- **Car Radar:** Higher detail (direction, closing speed, range). Best for multi-class endurance racing and race starts where closing speed differentials are large and you need spatial context.

You only need one active at a time. Toggle between them depending on the session type.

Track Map Display shows the full circuit with real-time car positions. It is most useful in two specific situations:

1. **Learning a new track:** On your first 5 laps of a new circuit, keep the map visible. It previews the next corner before you see it — left or right, tightening or opening, braking zone depth. After 10 laps, turn it off — you should be reading the track, not a map.
2. **Traffic management in endurance races:** When catching or being caught by traffic, the map shows you where slower cars are on the full circuit. You can plan passes one sector ahead: "There is a GT3 at the hairpin; I will catch them on the straight after."

Finding time: The map shows sector splits. If you are losing time in sector 2 consistently, focus your next practice session entirely on sector 2 corners. Do not waste laps repeating sectors where you are already fast.

SubStanding is an F1-style live timing tower. Position, gap to leader, gap to car ahead/behind, fastest lap marker, and pit stop indicators.

How to use:

- **Purple sector:** Someone ahead just set the fastest sector time. Watch who – if it is the car you are chasing, they found pace. If it is the car behind you, they are coming.
- **Pit indicators:** When cars ahead show a pit icon, their stop time appears. If they lose 25 seconds in the pits and you are 15 seconds behind, you will jump them with your own stop – track position gained.
- **Fastest lap tracking:** If you are out of the podium battle but the fastest lap bonus is available (1 point in many series), and you have a 10-second gap both ahead and behind, pit for fresh tyres on the penultimate lap and go for the fastest lap.

The Setup Market lets you browse and download car setups shared by other drivers through an in-game app.

How to use:

1. Open the app in a practice session. It shows setups filtered for the car and track you are currently on.
2. Sort by rating (most upvoted) or by lap time (fastest).
3. Download the top 3 setups for your combination.
4. **Load setup 1, do 3 laps, note the feel.** Load setup 2, do 3 laps, note the difference. Load setup 3, do 3 laps. Pick the one that feels most stable and suits your driving style.
5. **Do not blindly trust lap times.** The top setup may have been set by an alien with an aggressive, unstable balance that suits their precision but will spin you out. Stability is faster than peak pace if the peak pace setup is undriveable.

How to learn from shared setups:

- Compare the downloaded setup to the default. Which parameters changed? More negative camber? Softer rear ARB? Lower tyre pressures? Each difference is a lesson in what that setup creator values.
- After a month of using downloaded setups, try building your own. Start from default, change one parameter at a time, and test. This is how you develop setup engineering intuition.

CompEditor is a standalone tool for creating and editing custom championships visually, without editing JSON by hand.

How to use:

1. Open CompEditor and create a new championship file.
2. **Add rounds:** Select tracks (order them to create a narrative flow – start with a track you know well, put the hardest circuit as the finale).
3. **Add cars:** Pick your grid. You can mix classes (e.g., GT3 + GTE + GT4 for multi-class) or run a single spec series.
4. **Configure AI:** Set AI driver names, skill levels (95–100% for a real challenge), and aggression (85–95% for clean racing). Name them after actual rivals or friends for added immersion.
5. **Points system:** Choose a preset (F1 25-18-15..., MotoGP 25-20-16..., or custom). For short seasons (5–8 rounds), use a tighter points spread so the championship stays alive longer.
6. **Weather:** Set weather per round. A wet race at round 3 and a dry finale creates an unpredictable championship arc. Random weather adds realism but can feel unfair – your call.
7. Save to `Documents/Assetto Corsa/champs/` and load from CM's Championship mode.

Design tip: A good championship tells a story. Round 1 is your home track (comfortable). Round 4 introduces a weather variable. The final round is a long endurance event (e.g., 45 minutes at Le Mans or Nordschleife). The narrative keeps you engaged across a season.

ForceDx is a DirectX FFB tuner that adds per-car filter curves and lets you bypass AC's built-in FFB post-processing.

When to use:

- Certain mod cars have dead or weak FFB compared to Kunos cars. ForceDx can amplify specific frequency ranges to restore feel.
- If your wheel feels "numb" on centre (no self-aligning torque through the steering rack), ForceDx adds a subtle centring effect at low speeds.
- If FFB clips (the wheel goes light when cornering forces exceed the motor's capability), ForceDx compresses the dynamic range so strong forces come through without losing detail.

Setup:

1. In Content Manager, go to **Settings** → **Assetto Corsa** → **Controls** → **Force Feedback**. Disable all post-processing effects (minimum force = 0%, kerb/road/slip effects = 0%).
2. Launch ForceDx. It will detect your wheel. Load the per-car profile for your current car.
3. Drive 3 laps. Adjust the **Gain** slider until you feel clear cornering forces without clipping on your wheel's FFB meter.
4. Save the profile. It auto-loads next session.

Signs you need ForceDx: FFB clipping indicator (in CM) is red in corners. The wheel goes completely light mid-corner. Mod cars have no self-centring force. These issues are per-car, not global – ForceDx fixes them individually.

Esotic AI Line Helper is a standalone tool for creating or repairing AI racing lines on track mods. Use it when AI cars crash at a specific corner, drive off track, or are unrealistically slow on a downloaded circuit.

How to use:

1. Open the tool and load the track's AI folder (inside the track's `ai/` directory).
2. The tool visualises the AI racing line, pit lane entry/exit, and fast/slow AI paths.
3. **If AI consistently crashes at Turn 3:** Open the AI line for that turn. The line likely cuts the apex too aggressively or runs wide on exit. Drag the spline to smooth it.
4. **If AI is slow on straights:** The AI fast path may have unnecessary braking points. Remove mid-straight brake markers.
5. Save the file. Test with a quick AI race at that track. The improvement should be immediate.

Pro tip: For popular track mods, check the RaceDepartment comments before creating your own AI lines – someone may have already uploaded a fixed `ai` folder.

4.11 Setup Influence on Driving Technique

At this level, your driving technique should inform your setup, not the other way around.

- **If the car understeers:** Try adjusting trail braking depth before changing the front ARB
- **If the car is unstable on exit:** Change your throttle application technique before softening the rear ARB
- **If the car does not rotate on entry:** Experiment with brake bias (more rear) before changing the suspension

The rule: **Drive around the problem first. If you cannot fix it with technique in 5 laps, then change the setup.**

Practice Focus: Phase 4

- **Cars:** Ferrari 488 GTE (GT3/GTE), any Formula car (open-wheel), mix of drivetrains
- **Tracks:** Nürburgring Nordschleife, Spa, Bathurst — challenging circuits with varied corner types
- **Conditions:** AI races (higher aggression), online practice, mixed weather (Pure)
- **Goal:** Race the AI with 95% aggression for 15+ lap stints. Then review telemetry. Then race again

The Complete Curriculum Path

For reference, here is the full Assetto Maximus Driving Curriculum:

Phase	Title	Wave	Focus
1	Foundations	Wave 0	Consistency, confidence, basic car control
2	Balance & Speed	Wave 1	Weight transfer, trail braking, slip angle
3	Cornering Precision	Wave 1	Advanced techniques, complex corners
4	Race Craft & Mastery	Wave 2	Overtaking, racecraft, mental game

Installation Sequence

1. Ensure Wave 1 is fully installed and verified
2. Install telemetry and data tools first (they integrate with CSP)
3. Install all content mods in any order
4. Configure MoTeC plugin in CM: **Settings** → **CSP** → **General** → **Enable telemetry logging**
5. If using Pure, verify it is active as the weather controller

Graphics Mods

This section catalogs all graphics-enhancing mods in Assetto Maximus. Graphics mods are available from Wave 1 onward.

Wave 1 Graphics

Weather & Lighting

- Pure — Advanced lighting, weather, and sky controller. Physically-based sky rendering, volumetric clouds, dynamic exposure, and realistic sun positioning. Delivers the most photorealistic lighting available for AC. Pure is the successor to the older Sol weather system and is now the recommended weather controller for all CSP installations. **Dependencies: CSP 0.2.0+**. **Impact: Lighting framework replacement — controls sky rendering, cloud scattering, and light dispersion. System: Weather/Lighting engine**.
- Custom Shaders Patch — Framework that enables Pure and other visual mods. Adds per-pixel lighting, dynamic shadows, rain effects, and post-processing. **Impact: Foundational graphics framework**. Use the latest stable build compatible with AC v1.16.4.

Pure vs Sol (legacy): Pure is the current standard. Sol was the predecessor and remains available for users who cannot upgrade to Pure, but Pure offers more realistic sky colors, cloud scattering, and light dispersion. Pure can coexist with Sol's weather plan files but should be the active controller.

Post-Processing Filters

Post-processing filters (PP filters) control color grading, tone mapping, and bloom — the final image "look." The right PP filter transforms AC's visuals from dated to photorealistic.

- Revolution PP Filter — Clean, neutral post-processing. **Dependencies: CSP, Pure**. **Impact: Visual tone mapping and color grading**. **Note: Replaces the old Reasonable PP Filter which is no longer available on OverTake**.gg.
- C13 Aegis PPFilter — Photorealistic post-processing filter. Designed for Pure/CSP. Features realistic color science, natural highlights, deep contrast. One of the top choices for photorealism. **Dependencies: CSP, Pure recommended**.
- A3PP (A3 Post Processing) — Dramatic, film-style PP filter. Warm tones, cinematic contrast, natural bloom. Popular for screenshots and racing. **Dependencies: CSP**.
- ILCB Photo Realistic Filter — PP filter specifically focused on photographic realism. Neutral color palette, subtle bloom, natural gamma. **Dependencies: CSP**.
- Natural PP Filter — PP filter designed for accuracy. Minimal artistic bias, clean highlights, natural shadow detail. **Dependencies: CSP**.

Choosing a PP filter: Try 2—3 filters and pick the one that looks best on your monitor. Filters affect different monitors differently based on panel type (IPS, VA, OLED) and calibration.

Texture & Lighting Enhancements

- **4K Skin & Texture Packs** — High-resolution car skins and track textures. Search OverTake AC Skins or browse specific packs: Porsche 911 RSR (WEC 2019, Le Mans 2020), Audi RS5 DTM, RSS Formula Hybrid Tyres. **Dependencies: Content Manager**.

4K Texture Packs

These packs replace stock textures across the game. Install in order: track surfaces first (biggest impact), then environmental, then car details. All require **Custom Shaders Patch** and are **Wave 1**.

Track Surfaces & Circuit Environment

- **4K Track Surfaces** — High-resolution asphalt, concrete, and kerb textures for Kunos circuits. Replaces the muddy default surfaces with sharp, photorealistic road detail. Single biggest visual upgrade for circuit driving — everything you look at is road. Search "4K track surfaces AC" on RaceDepartment for the most current pack.
- **4K Grass Textures** — Replaces the default blotchy green grass with detailed, varied grass textures. Multiple biome variants available (European, desert, autumn). Search RaceDepartment for the latest version compatible with CSP.
- **4K Tree & Foliage Textures** — High-res tree bark, leaf, and bush textures. Makes forested circuits (Nordschleife, Road America, VIR) look current-generation instead of 2014-era. Search RaceDepartment for "4K tree textures AC".
- **4K Kerbs & Rumble Strips** — Detailed kerb textures with visible aggregate, painted colour bands, and wear marks. The red-and-white rumble strips finally look like real concrete instead of blurry blocks.

Sky & Weather Textures

- **4K Sky & Cloud Textures** — High-resolution skybox textures with detailed cloud formations, sunset gradients, and star fields. Replaces the low-res vanilla sky that shows pixelation at high FOV. Compatible with Pure and Sol weather engines.
- **Rain & Spray FX** — Updated raindrop textures on the windscreen and spray particle effects. Works with CSP's rain implementation. Visible droplets instead of blurry smears.

Car Interior & Garage

- **Interior Textures** — Replace low-res dashboard, steering wheel, and seat textures for Kunos cars. Gauges are readable, carbon fibre weave is visible, and alcantara looks like fabric, not plastic.
- **Showroom & Garage Textures** — Updates the CM showroom environment textures. Your previews and garage views match the quality of the in-game assets.

Priority Order

Install in this order for maximum visual impact per gigabyte downloaded:

1. **Track Surfaces** (visible on every corner of every lap)
2. **Grass + Trees** (visible on every circuit with foliage)
3. **Sky & Clouds** (visible in every session, especially with Pure)
4. **Kerbs** (visible at every apex and track-out)
5. **Rain & Spray** (only when using CSP rain)
6. **Interiors** (only visible in cockpit cam)
7. **Showroom** (only visible in menus)

Optional: Reshade

- Reshade — External post-processing injector. Adds ambient light, bloom, SMAA, LUT-based color grading, and dozens of other effects on top of AC/CSP output. **Dependencies: Standalone (injects into AC process)**. **Impact: Additional post-processing layer. Can conflict with some CSP features — test per-filter**.

Note: Reshade is powerful but adds GPU overhead (5—15% frame time impact). Many users find Pure + C13 Aegis sufficient without Reshade.

Photorealism Checklist

To achieve a photorealistic look:

1. Install CSP (foundation)
2. Install Pure (lighting/weather)
3. Install a PP filter — C13 Aegis or A3PP recommended
4. Track textures: ensure tracks have hi-res surface textures
5. Configure CSP Lighting FX: enable dynamic exposure, high-res reflections
6. Configure Pure weather controller for your region and time of day
7. (Optional) Reshade for final color grading if desired
8. Test PP filter selection — visual preference is subjective

Content Mods

This section catalogs all car and track mods in Assetto Maximus. Content mods are available from Wave 1 onward.

Mod entries follow this format: **Name** — Description. **Dependencies**. **Compatibility notes**.

Cars

Wave 1 Cars — Spec & Open Wheel

- Mazda MX-5 Cup 2016 — Spec racing series car. Perfect for learning racecraft. **Dependencies: None**.
- Caterham Seven Academy 2017 — Two variants (LDH/RHD). Open-top spec racer with no assists. Excellent for building car control fundamentals.
- MINI Cooper JCW Challenge (F56) 2019 — Modern FWD spec-racer. Teaches front-wheel-drive dynamics and trail-braking.
- Skip Barber F2000 — Tube-frame open-wheeler. The classic American racing school car — low power, high grip, teaches momentum driving.
- Dallara F312/F317 (Formula 3) — Two cars covering the 2012 and 2017 F3 eras. High-downforce single-seaters for your first taste of proper downforce driving.

Wave 1 Cars — Touring & GT

- Nissan Primera BTCC 1999 — Super Touring-era icon. Front-wheel-drive touring car with period-correct aero and racing pedigree.
- BMW E36 320i STW-BTCC 1998 — RWD touring car from the golden Super Touring era. Natural rival to the Primera.
- SEAT León TCR 2018 — Modern FWD touring car. TCR-spec aero and slicks with production-derived chassis.
- Ferrari 488 GTE — GTE/GTLM endurance racer. Community conversion based on technical documentation. **Dependencies: None**.
- Bentley Continental GT3 — Thunderous GT3 racer. Massive presence on track, unique exhaust note.
- Caterham Roadster 500 (fictional) — A 500 bhp/tonne monster. Takes the Seven formula to its extreme limit.

Wave 1 Cars — Classics & Period

- Abarth 1000 TC Berlina Corsa 1966 — Four variants (Stock/Gr2/Gr5/Slalom). Tiny 1-litre rear-engined racer. Teaches momentum conservation on a shoestring budget.
- BMW E30 325i Touring 1987 — Two variants (RWD/AWD). The classic boxy BMW wagon, ideal for casual track days and Nordschleife laps.
- TVR Griffith 200 1963 — Lightweight British V8 sports car. No ABS, no TC — raw analog driving.

- Alfa Romeo Giulia 1750 GTAM 1970 — The racing evolution of the Giulia sedan. 220 bhp, 920 kg, and a screaming twin-cam that defined an era. Teaches throttle-steering in a lightweight FR chassis.
- Alfa Romeo Giulia TZ 1963 — Two variants (Stock/Competition). Zagato-bodied homologation special. Tiny, low-drag, and rev-happy. Teaches carrying minimum speed through corners.
- Porsche 356A Carrera GT 1958 — Two variants (stock/SCCA). 4-cam Fuhrmann engine in a 850 kg body. The car that launched Porsche's motorsport legacy. Rear-engine pendulum effect teaches weight transfer management.
- Porsche 550/1500 RS Spyder 1953 — The original mid-engine Porsche racer. 110 bhp, 550 kg, and absolutely telepathic steering. Teaches mid-engine corner entry rotation.

Wave 1 Cars — Modern Performance

- Ginetta G55 GT4 Supercup 2017 — Two variants (LDH/RHD). V6-powered British GT4 car. The sweet spot between road car and full GT3 — enough aero to feel it, not enough to rely on it.
- Honda NSX 1990 — Four variants (NA1, Type R, Acura, Acura S1). The everyday supercar. Mid-engine, VTEC, and chassis by Senna. Teaches mid-engine throttle control at approachable speeds.
- Ferrari 512 TR 1991 — Two variants (stock/S1). Flat-12 Testarossa successor. The last of the analogue V12 Ferraris. Heavy, powerful, demanding — teaches brake management in a heavy mid-engine car.
- Koenigsegg One:1 — 1,360 bhp, 1,360 kg. The 1:1 power-to-weight hypercar. A physics exercise — too much power for any corner, too much aero for any excuse. Teaches throttle discipline at the extreme.
- Corvette Daytona Prototype 2014 — Two variants (Dallara/Coyote chassis). IMSA prototype with Corvette bodywork and 600 bhp V8. The bridge between GT cars and full prototypes.
- Citroën DS3 WRC — 1.6L turbo, 4WD, 380 bhp. Sebastien Loeb's weapon of choice. Teaches left-foot braking, Scandinavian flicks, and loose-surface car control.
- Formula Student UCM 2016 — University-built single-seater. 80 bhp, 220 kg, massive wings for its weight. The accessibility of a go-kart with the aero sensitivity of an F1 car. Teaches downforce driving at low speed.

Car Packs Worth Getting

- VW Polo R WRC — Part of the RallyLegends series of WRC and rallycross cars (search for all entries by UltraNew_B on RaceDepartment).
- **Caterham Seven range** — Search RaceDepartment for "Caterham Seven" — there are 165, Academy, 420R, and Roadster variants by different authors that form a near-complete Caterham ladder.
- BTCC / Super Touring — Search RaceDepartment for "BTCC" — multiple 1990s Super Touring cars exist (Volvo, Vauxhall, Renault, Honda).

Wave 2 Cars — Prototypes & Formula

These are unlocked after completing Wave 2 fundamentals. High downforce, high power, high commitment. If you can drive these at the limit for a full stint, you can drive anything.

- McLaren MP4/4 1988 — The most dominant F1 car in history. 1.5L turbo V6, 650 bhp in qualifying trim, massive rear grip from ground-effect tunnels. Teaches you to trust aero and brake impossibly late. **Dependencies: None**. **Note: 1.3 — community conversion, not an official RSS/VRC car**.
- Williams FW31 2009 — 2.4L V8 F1 car from the last year of the V8 era. 18,000 rpm scream, 750 bhp, KERS boost button. Teaches hybrid-era energy management. **Dependencies: None**.

- Williams FW37 2015 (VRC) — Professional-grade F1 mod by Virtual Racing Cars. 1.6L turbo hybrid with full ERS deployment strategy. The closest you'll get to modern F1 in AC. **Dependencies: None**.
- Oreca FLM09 Prototype Challenge — 450 bhp spec-prototype for endurance racing. Open cockpit, high downforce, sequential gearbox. The entry point to prototype driving — fast enough to demand respect, forgiving enough to learn. **Dependencies: None**.
- Toyota TS020 GT-One 1998 — Legendary LMP. 3.6L twin-turbo V8, 600+ bhp, Le Mans icon. Massive speed difference to GT traffic requires disciplined traffic management.

Wave 2 Cars — GT Endurance & DTM

- Opel Calibra V6 DTM 1996 — 4×4 DTM monster. 500 bhp V6, advanced aero, and all-wheel-drive traction through the corners. Completely different driving style to RWD touring cars — you steer with the throttle. **Dependencies: None**.
- Caterham Seven 420R — The top of the Caterham ladder. 210 bhp in 560 kg = 375 bhp/tonne. Slicks, sequential gearbox, and aero. Demands absolute precision. **Dependencies: None**.
- Nissan Skyline R33 GT-R LM 1995 — Le Mans homologation special. 400+ bhp RB26 with ATTESA AWD. Heavy on corner entry, rocketship on corner exit. Teaches momentum management in a heavy car.

Wave 2 Cars — Unrestricted Road Cars

- Dodge Viper RT/10 1992 — 8.0L V10, 400 bhp, no ABS, no TC, no mercy. The ultimate test of throttle discipline. Gets you ready for the kind of power that punishes mistakes instantly.
- Bugatti Type 32 1923 — The "Tank." 2.0L straight-8, drum brakes, 100-year-old chassis flex. If you can set a clean lap in this, modern cars will feel like cheating.

Car Mod Requirements

All car mods must:

- Be in `.zip`, `.rar`, or `.7z` format
- Contain a `ui_car.json` or be recognized by CM
- Place files in `assettocorsa/content/cars/[car-folder]/`
- Work with AC v1.16.4 physics engine

Tracks

Wave 1 Tracks — Real Circuits

- Mount Panorama Bathurst — 6.2 km, 23 turns. Public road turned race track. Features the iconic Conrod Straight and terrifying downhill section. **Dependencies: None**.
- Nürburgring Nordschleife — 20.8 km, 154 turns. The ultimate driver's challenge. **Dependencies: None**. **Note: This is a community conversion; verify compatibility with AC v1.16.4**.
- Spa-Francorchamps — 7.0 km, 19 turns. 2022 layout update. **Dependencies: None**.
- Brands Hatch — 3.9 km, 11 turns. Classic British circuit. **Dependencies: None**.
- Silverstone Circuit — 5.9 km, 18 turns. FIA Grade 1. 2024 layout. **Dependencies: None**.
- Monza — 5.8 km, 11 turns. Temple of Speed. 2022 update requires CSP. **Dependencies: CSP**.
- Road America — 6.5 km, 14 turns. American road racing cathedral. Deep braking zones and sweeping curves through Wisconsin forests.

- Road Atlanta — 4.1 km, 12 turns. Home of Petit Le Mans. The Esses sequence demands commitment.
- Sebring International Raceway — 6.0 km, 17 turns. Legendary bumpy airfield circuit. Tests suspension and durability.
- Riverside International Raceway — Four layouts. Classic California road course, now lost to history.
- Sandown Raceway — 3.1 km, 13 turns. Australian V8 Supercars venue. Short, fast, unforgiving.
- Sachsenring GP — 3.7 km, 14 turns. German MotoGP circuit. Tight, flowing, elevation-heavy.

Wave 1 Tracks — Freeroam & Touge

- LA Canyons — 64 km of canyon roads in the San Gabriel Mountains. Freeroam servers, drift spots, and the famous Snake section. The definitive AC cruising map.
- Hakone Nanamagari Touge — Classic Japanese mountain pass. Narrow, technical, cambered corners — perfect for touge battles.
- Hakone Ohiradai Touge — Upper course of Hakone. Faster and more open than Nanamagari.

Wave 1 Tracks — Rallycross & Test

- Holjes Rallycross — The "Magic Weekend" venue in Sweden. Mixed-surface RX track with huge elevation changes and the iconic Velodrome corner.
- Karelia Cross — Finnish-style rallycross circuit. Gravel and asphalt mix.
- Skidpad — Circular test track with markings. Essential for testing car behavior, drift angles, and tire limits.

Wave 1 Tracks — More Real Circuits

- Donington Park 2018 — GP + National layouts. 4.0 km, 12 turns. Flowing British parkland circuit. The Craner Curves and Old Hairpin are among the most satisfying corner sequences in motorsport. Famous for Senna's 1993 European GP lap-of-the-gods.
- Watkins Glen — 5.5 km, 11 turns. Classic American road course. The Esses (T2–T4) are taken flat in most cars — a commitment test. The Boot and Outer Loop demand late apexes to set up the long back straight.
- Circuit Gilles Villeneuve (Montreal) — 4.4 km, 14 turns. Semi-street circuit on an island in the St. Lawrence River. The Wall of Champions awaits the overconfident. 300 km/h back straight into a tight chicane.
- VIR — Virginia International Raceway — 5.3 km, 17 turns. The American Nordschleife. Fast esses, blind crests, and the intimidating Climbing Esses. Zero runoff in places.
- Okayama International Circuit — 3 layouts. 3.7 km, 11 turns. Former Pacific GP venue. Tight, technical infield followed by two long straights. Track position defines the race.
- Paul Ricard / Le Castellet — 4 layouts. 5.8 km. Modern F1 test venue with distinctive blue and red runoff. 1.8 km Mistral straight. Signes corner is flat in GT cars with proper downforce.

Wave 1 Tracks — Short & Technical

- Cadwell Park — 3.5 km, 11 turns. The "Mini Nürburgring" of England. Narrow, undulating, zero runoff. The Mountain is a blind crest where cars get airborne. Pure commitment circuit.
- Goodwood Circuit — 3.8 km, 7 turns. Historic 1948 airfield circuit. Fast, flowing, unchanged since the 1950s. Perfect for classic car events and Revival meeting atmosphere.
- Norisring 2009 — 2.3 km, 8 turns. The Monaco of Germany. A DTM street circuit. Four 90-degree corners linked by long straights on public roads. Simplest layout, hardest to master.

- Highlands Motorsport Park — 4.1 km, New Zealand. Modern circuit in the Southern Alps. The Bridge to Nowhere and a Corkscrew-like descent. Stunning scenery with punishing elevation.
- Brno Masaryk Circuit — 5.4 km, Czech Republic. Former MotoGP venue with wide, sweeping corners and big elevation. The horsepower circuit — carries speed everywhere.

Wave 1 Tracks — Japanese & Touge

- Ebisu Higashi (East) — 2.1 km Japanese touge/circuit hybrid. The spiritual home of drifting. Tight, technical, camber-heavy corners reward aggressive rotation.
- Ebisu Kita (North) — The north course of Ebisu. Faster and more open than Higashi. The jump section and legendary downhill sweeper test suspension composure.
- Otarumi Touge — Long Japanese mountain pass. The most flowing of the 90s GDP touge series. Sweeping corners, tunnel sections, and constant elevation drops.
- Hero Shinoi — 4.1 km Japanese technical circuit. A hidden gem. Fast esses, tight hairpin, and elevation changes rivalling Spa. Best for hotlapping and online battles.

Wave 1 Tracks — Open Roads & Classics

- Alto Tajo Hillclimb, Spain — 8.5 km point-to-point mountain climb through the rural Taravilla region. Narrow asphalt, blind crests, and stone walls. Requires total commitment — one mistake means a 50-metre drop. For hillclimb events and time attack. **Note: Point-to-point; use with time trial or practice mode. No pit boxes**.
- Toscana Autodromo — Tuscan countryside circuit set among rolling hills. Wide, flowing layout with fast sweepers and a long elevation sequence through olive groves. Designed for high-speed GT and prototype testing. **Dependencies: None**.
- Trial Mountain — Gran Turismo classic faithfully recreated. 5.5 km fictional circuit with a steep uphill section, downhill tunnel double-apex, and the iconic bridge jump. Corners have names from the GT era — Carousel, Hairpin, the Esses. Perfect nostalgia lap for anyone who grew up on GT1–4. **Dependencies: None**.
- Aosta Grand Prix — 16 layouts in one package. Based on a real Italian region, this versatile pack includes a full GP circuit, drift layout, oval, wet weather road course, and light/mod variants. The swiss army knife of track mods — one download covers everything from drift practice to endurance racing. **Dependencies: None**. **Note: 16 layouts in CM; collapse the group to avoid clutter**.

Wave 2 Tracks — Endurance & Long Circuits

These are longer, more demanding circuits for extended sessions, multi-class racing, and stamina events. Best experienced after mastering Wave 1 tracks.

- Nordschleife Endurance + Cup — The full combined VLN layout (Nordschleife + GP circuit). 25.9 km with expanded 60-car pit boxes. The definitive endurance racing venue. **Dependencies: Nordschleife mod installed**.
- Circuit de la Sarthe (Le Mans 24h) — 13.6 km of public roads and permanent circuit. Mulsanne Straight, Indianapolis, Porsche Curves. Requires patience, top-speed setup, and the ability to manage 350+ km/h closing rates in LMP vs. GT traffic.
- Daytona International Speedway — Road Course — 5.7 km Roval. The Bus Stop chicane, high-banked oval sections, and infield technical section. Home of the Rolex 24.
- Fuji Speedway — 4.5 km, 16 turns. The 1.5 km front straight is one of the longest in motorsport. Heavy braking into Turn 1 sets up the entire lap. Former F1 venue, current WEC staple.

- Suzuka Circuit — 5.8 km, 18 turns. Figure-8 layout with the iconic Esses, Degner curves, and 130R. Widely considered the ultimate driver's circuit alongside the Nordschleife. Every corner type represented on one lap.
- Targa Florio — 72 km of Sicilian mountain roads through the Madonie range. The historic 1906–1977 endurance race route, faithfully recreated. Over 800 corners, villages you drive through, stone walls inches from the road, and elevation changes from sea level to 600 metres. Three layouts: **Full Circuit** (72 km, 35 minute laps in a GT3), **Short Course** (44 km), and **Medium Course** (55 km). This is not a race — this is a pilgrimage. Bring a car you trust, turn off the HUD, and drive. **Dependencies: None**. **Note: The full circuit has no pit boxes — use the shorter layouts for racing. The full layout is for hotlapping and experiencing one of motorsport's greatest lost circuits**.

Track Mod Requirements

All track mods must:

- Contain valid surfaces and AI lines
- Be compatible with AC v1.16.4 track structure
- Include at least one pit box layout
- Install to `assettocorsa/content/tracks/[track-folder]/`

Freeroam & Traffic

Freeroam tracks like LA Canyons and mountain passes are beautiful but empty without traffic. These mods populate the roads with AI vehicles, transforming cruising maps into living environments.

- **2REAL Traffic Simulation** — CSP-based traffic system that spawns AI-controlled civilian vehicles on freeroam maps. Cars follow road rules, stop at intersections, signal turns, and react to your presence. Configurable: traffic density (light rush hour to empty midnight), vehicle mix (economy cars to trucks), and driver aggression (Sunday drivers to Sicilian). Works with LA Canyons, Shutoko Revival Project, Union Island, Pacific Coast Highway, and most other freeroam maps. **Dependencies: CSP 0.1.79+, Content Manager**. **Install: via CM → Content → Mods**. **Wave 1**. **Note: Enable "CSP Traffic" in CSP settings after installing. Each freeroam map needs a traffic layout — most popular maps include one**.
- **Shutoko Revival Project (SRP)** — 160+ km of Tokyo expressway network, faithfully recreated. The ultimate night-cruising and highway racing environment. Multiple interchanges, tunnels, and the iconic C1 inner loop. Includes its own traffic system (separate from 2REAL). **Dependencies: CSP**. **Wave 2**. **Note: 8+ GB download. Best experienced in VR or triples. Use with Pure for rain-soaked expressway atmospherics**.

Choosing Freeroam Maps + Traffic

- For **daytime canyon/mountain driving**: pair LA Canyons or Pacific Coast Highway with 2REAL Traffic.
- For **nighttime highway racing**: Shutoko Revival Project has its own built-in traffic — 2REAL is not needed.
- For **touge battles**: hakone tracks and touge passes are best without traffic. Traffic adds atmosphere for cruising, not racing.

Career & Campaign

Assetto Corsa’s built-in career mode is limited. These mods add proper progression systems — championships, economy, rival AI, and structured career paths.

Overview

	RS: Careers	RS: Y2K	ApexRivals LITE
Author	noonecares11	noonecares11	EverlastingApex
Version	1.6.1	1.4	1.0.19b
Released	Jun 2022	Mar 2026	Jul 2024
Status	Stable	Active dev	Active dev
Price	Free	Free	Free (LITE)paid upgrade
Reviews	170	14	5
Cars	44 (vanilla+DLC)	External mod cars	Your own car list
Economy	Yes	Yes	No (Elo only)
Persistent damage	No	Yes	No
Freeroam	No	Yes	No

Table 3: Career mod comparison

Redline Stars: Careers Lite

Redline Stars: Careers — 32 championships across 44 cars, with sponsorships, bank loans, teammate battles, post-race interviews, and a social media simulation. You play as a dentist with a midlife crisis.

Dependencies: AC Ultimate Edition (all DLC). **Compatibility**: Uses the vanilla AC launcher (not CM) — you must have fewer than 450 cars installed. Free, no ads or IAP. 18+ language. Fully custom championships with mod car/track support, cloud saves, adjustable difficulty, random weather, and ballast/engine restrictor balancing.

Redline Stars: Y2K Lite

Redline Stars: Y2K — The sequel set in the year 2000. License tests, freeroam street racing, drift challenges, car auctions, a paint shop, and deep tuning (cams, flywheels, brakes, weight reduction). Features persistent damage (crashes cost money), car maintenance (engine condition degrades), and a dynamic economy where prices fluctuate by age/mileage/condition, culminating in the 2008 financial crisis.

Dependencies: External Gumroad car mods and track mods (not included). **Compatibility**: Active development — expected to eventually supersede RS: Careers. These two mods should not be installed together (conflicting save structures). Free with optional Patreon support.

ApexRivals Career Mode LITE Lite

ApexRivals Career Mode LITE — Not a traditional career. Pure Elo-based league system with persistent AI rivals who continue racing and progressing even when you're offline. You and the AI follow the same rules for promotions and team offers. Supports both Assetto Corsa and Automobilista 2.

Dependencies: None. **Compatibility**: Runs as a standalone external app (not a CM career folder) — compatible alongside any of the above career mods. The LITE version lacks multiclass racing and long seasons (unlocked in the paid full version). Built-in SOL & PURE weather support.

Content Manager Championships

Content Manager includes a built-in **Championship** mode (CM → Championship) that lets you create custom multi-race series with:

- Custom car and track selection
- Points systems (F1-style, GT-style, or custom)
- Practice / Qualifying / Race sessions
- AI opponents at configurable skill levels
- Weather progression across rounds

No additional mod needed — this is a CM feature that many players overlook. Available immediately from Wave 0.

Choosing a Career Mod

- For the most complete experience with the least hassle: install **RS: Careers** (one download, all content included, 32 championships).
- For a living, breathing car world with damage, freeroam, and a GT2-style vibe: install **RS: Y2K** (still in early access but already deeper than any competitor).
- For competitive league racing without economy fluff, and if you also own AMS2: install **ApexRivals** alongside either RS mod.
- Avoid installing RS: Careers and RS: Y2K together — they conflict. ApexRivals can coexist with either.

Apps, Sounds & Resources

Beyond cars and tracks, Assetto Corsa has a rich ecosystem of apps, utilities, sound mods, and community resources that enhance every session. These mostly come from the community mod spreadsheet maintained by Epistolarius (2015–2020).

Essential Apps

All of these install to `apps/python/[app-folder]/` and can be toggled on/off from the in-game app bar.

- **Helicorsa** — Radar-style proximity indicator showing cars alongside and behind you. Essential for clean wheel-to-wheel racing. Shows distance and closing speed. **Wave 0**.
- **Track Map Display** — Full circuit map overlaid with live car positions. Useful for learning track layouts and monitoring traffic during longer races. **Wave 0**.
- **SubStanding** — Live leaderboard with gaps and sector times, similar to F1 TV graphics. Shows position changes, fastest laps, and stint analysis. **Wave 0**.

Weather & Visuals

- **Pure** — Advanced lighting, weather, and sky controller. Physically-based sky rendering, volumetric clouds, dynamic exposure, and realistic sun positioning. Successor to the older Sol weather system. Delivers the most photorealistic lighting available for AC. Required by multiple career mods and all serious weather-dependent driving. **Dependencies: CSP 0.2.0+**. **Wave 0**.
- **Sol (legacy)** — Predecessor to Pure. Still functional and available for users who cannot upgrade to CSP 0.2.0+. **Dependencies: CSP**. **Note: Pure is strongly recommended over Sol for all new installations**.
- **Out Run Weathers** — Warm and cold weather presets designed for the Out Run filter. Gives coastal sunset and crisp winter morning vibes. **Wave 1**.
- **Black Pearl PP Filter** — Post-processing preset for both in-game sessions and CM preview creation. Neutral, car-showroom look. **Wave 1**.

Track Enhancements

- **Autumn Textures Pack** — Replaces summer foliage with autumn colours for Nordschleife, Imola, Monza, Mugello, Spa, and others. Perfect for fall-themed championships. **Wave 1**.
- **RBS — Real Billboards & Stuff** — Replaces track-side billboards with real-world sponsors and adds winter textures for Monza and Nordschleife. **Wave 1**.
- **Kunos Tracks Extra Pits** — Expands stock circuits to 32–40 pit boxes. Nordschleife Endurance version supports 60 cars. **Dependencies: None**. **Note: Clutters track selection in CM; install only what you need**.

Skins & Textures

Texture and skin mods replace visual elements without affecting physics or performance. They install directly into track and car folders.

- **Pit Entry Markers** — Adds clear visual markers and speed signs at pit entry for all Kunos tracks. Removes the guesswork — no more accidental pit lane speeding penalties. Original files untouched, no checksum issues. **Wave 0**.

- **GT-Style Race Flags** — Replaces the stock flags in the top-left corner with GT-style indicators. Yellow, blue, white, and checkered flags are cleaner and more visible during races. **Wave 0**.
- **TK's Smoke Mod** — Redesigned smoke and dust particle effects. More realistic tyre smoke on lock-ups and burnouts, and better dust clouds on gravel traps and off-track excursions. **Wave 1**.
- **HD Windscreen Damage Texture** — Replaces the default glass crack texture with a higher-resolution, less intrusive version. The original texture and glass simulation (road car glass vs. race car plastic) isn't good — this texture is far less annoying while not sacrificing the effect. **Wave 1**.
- **Laguna Seca 2018 WeatherTech Textures** — Updates Laguna Seca signage and branding to the 2018 WeatherTech Raceway era. Pairs well with any modern GT or IMSA championship. **Wave 1**.
- **Black Cat County — Normal Roads** — Replaces the highway road texture on Kunos' fictional Black Cat County with standard asphalt. The original bright highway markings break immersion — this makes it feel like a proper circuit. **Wave 1**.
- **National Flag Icons — Blancpain GT Style** — Adds proper national flag icons in the Blancpain GT Series style for CM car previews and UI elements. Makes your grid look like a real GT World Challenge broadcast. **Wave 1**.
- **Brand Badges Pack** — High-quality manufacturer badge textures for CM car previews and UI. Covers dozens of marques. Essential if you want your car list to look professional instead of showing missing badge placeholders. **Wave 0**.

Telemetry & Setup

- **ACTI — AC Telemetry Interface** — Data logging and live telemetry with export to Motec i2 Pro analysis software. Better support for the latest AC shared memory than the official AiM integration. **Dependencies: None**. **Wave 2**.
- **The Setup Market** — Browse, share, and download car setups through an in-game app. Especially useful when learning a new car and needing a baseline tune. **Wave 1**.
- **Esotic AI Line Helper** — Standalone tool for creating and repairing AI racing lines on track mods. Essential if you install obscure tracks with broken or missing AI. **Wave 2**.

Pro Tools (Wave 2)

These are advanced tools for serious drivers. They provide data, feedback, and analysis capabilities that let you find tenths you didn't know existed.

- **Sidekick** — Real-time tire and brake temperature monitor with multi-lap comparison. Shows pressure build-up, graining risk, and optimal operating windows per compound. More detailed than the stock Kunos tire app. **Install:** `apps/python/Sidekick/`
- **Camber Extravaganza** — Live camber angle display for each tire, updated in real time through corners. Essential for dialing in suspension geometry — you can see exactly how much camber you lose to body roll at each corner. **Install:** `apps/python/CamberExtravaganza/` .
- **Car Radar** — Advanced successor to Helicorsa. Shows proximity in 360 degrees with closing speed vectors, blind-spot warnings, and configurable detection ranges. Essential for multi-class racing where closing speeds can exceed 100 km/h. **Install:** `apps/python/CarRadar/` .
- **Race Essentials** — Comprehensive race dashboard: fuel calculator, stint prediction, gap to ahead/behind, pit stop countdown, and tire life estimation. Replaces 4–5 individual apps with one clean overlay. **Install:** `apps/python/RaceEssentials/` .
- **ForceDx** — DirectX force feedback tuner. Lets you bypass AC's built-in FFB post-processing and apply per-car filter curves. Fixes dead FFB on certain mod cars and lets you tune feel independently of Kunos settings. **Dependencies: Requires DirectX wheel (not OSW/SimuCube)**. **Install: standalone** `.exe` .
- **CompEditor** — Edit championship files visually instead of editing JSON by hand. Add custom grids, modify points systems, set AI driver names, and configure weather for every round. **Install: standalone** `.exe` .

Sound Mods

Sound mods replace the `*.bank` FMOD audio files for specific cars. Always back up the original `.bank` file before installing.

- **Fonsecker Sound Mods** — The most prolific AC sound modder. High-quality engine sound replacements for dozens of Kunos cars. Search RaceDepartment for "Fonsecker" to find the full catalogue.
- **HD Windscreen Damage Texture** — Replaces the default glass crack texture with a higher-resolution, less intrusive version. Subtle but noticeable improvement. **Wave 1**.

Modding Resources

- **AC v1.16 Community Mod Spreadsheet** — The definitive curated list of community mods by Epistolarius. Covers cars, tracks, liveries, apps, sounds, and resources — all tested on AC v1.16.x. Use the tabs at the bottom to browse by category.
- **Corner Names Pack** — Adds real corner names to track maps for 50+ circuits. Great for learning corner-by-corner references.
- **Brands Hatch Pit Lane AI Fix** — Corrected `pit_lane.ai` file to prevent AI cars from crashing into pit entry walls. Check for similar fixes for other troubled circuits.

Wave Placement

- **Essential Apps**: Install in Wave 0 alongside CM — no mechanics impact.
- **Weather & Visuals**: Pure (or legacy Sol) requires CSP and is essential from Wave 0. PP filters and textures are Wave 1.
- **Telemetry & Tools**: Setup Market in Wave 1. ACTI and AI tools in Wave 2.
- **Sound Mods**: Install anytime. Back up original `.bank` files first.

Mechanics Mods

This section catalogs mods that alter or enhance physics, force feedback, and driving mechanics. These mods are available from Wave 1 onward and **never** in Wave 0.

Physics Mods

- FFB Clip — Analyzes force feedback output and prevents clipping (signal saturation). Provides real-time monitoring and auto-adjustment of FFB gain. **Dependencies: Content Manager System/ Mechanic Impact: Force feedback system enhancement. Does not change physics, but improves wheel feel and prevents signal distortion. Configure via FFB Clip UI or CM settings.**

- Minimalista HUD — Clean data HUD with tyre temps, brake bias, fuel, and delta times. **Dependencies: Content Manager**. **Impact: Information display. No physics changes**. **Note: Replaces the old RST Minimal UI**.

Tyre & Handling Mods

Future waves may include dedicated tyre models (e.g., Tyre RF Pro) or handling overlays. For Wave 1, the stock AC tyre model with CSP’s enhanced tyre FX is sufficient.

FFB & Wheel Settings

Detailed force feedback configuration can be found in the **Configuration** chapter.

Mod Compatibility Notes

- Mechanics mods must be tested together for conflicts
- FFB Clip works alongside all other mods
- HUD replacements like RST Minimal UI may conflict with other HUD mods — enable only one at a time
- No mechanics mod in Assetto Maximus modifies car performance data directly (no “better physics” cheat mods)

UI & Quality of Life Mods

This section catalogs all UI, HUD, and quality-of-life mods. These are the foundation of Wave 0 and remain active through all later waves.

Wave 0 (Core)

These mods are part of the Wave 0 installation and should be installed before any other mods.

Spotter & Audio

- Sidekick — Audible spotter that calls out: car proximity (left/right), gap distances, yellow/blue/checkered flags, pit lane entry/exit reminders. **Dependencies: Content Manager Impact: Audio. No visual overlay. No gameplay change. Reduces cognitive load by providing critical race information verbally. Configuration: Volume, language, and verbosity settings available in-app.**
- Crew Chief — External application providing: spotter calls, pit strategy advice, fuel calculations, tyre temperature readings, weather reports. **Dependencies: Standalone app (Windows). Runs alongside Assetto Corsa. Impact: External tool. Communicates via AC shared memory. No in-game modification. Setup: Install, launch before AC, select AC as the game. Voice recognition supports common commands.**

Visual Overlays

- Car Radar — 360° radar display positioned in the corner of the screen. Shows all nearby cars with distance and relative speed indicators. **Dependencies: Content Manager Impact: Visual overlay. No gameplay change. Essential for close-quarters racing awareness. Configuration: Radar range, size, opacity, and position.**
- Helicorsa — 3D arrow indicator drawn in-world above nearby cars. Arrow color and size indicate proximity. Green = safe distance, Yellow = close, Red = overlapping. **Dependencies: Content Manager Impact: Visual overlay. No gameplay change. Configuration: Trigger distance thresholds, arrow size, and enabled modes.**

Wave 1 Additions

- Minimalista HUD — Clean data HUD overlay. Replaces the old RST Minimal UI. **See 07-modlist-mechanics for full details.**
- 3D Shift Light — Visual gear indicator and shift light that displays optimal shift points. **Dependencies: Content Manager Impact: Visual overlay. No gameplay change.**

UI Mod Conflicts & Rules

- Do not enable two HUD-replacement overlays simultaneously
- Sidekick and Crew Chief can run together safely (Crew Chief defers to Sidekick for spotter calls when configured)
- Car Radar and Helicorsa complement each other — running both is recommended
- All Wave 0 UI mods are compatible with all later wave mods

Configuration Guides

Graphics Settings

Video Settings (in-game)

Setting	Recommended Value	Notes
Resolution	Native monitor resolution	Match your display
VSync	Off	Reduces input lag
Fullscreen	On	Better performance
Anti-aliasing	4x MSAA	Balance quality/performance
Anisotropic Filtering	16x	Minimal performance cost
World Detail	High	
Reflection Quality	Low to Medium	Heavy performance impact at High
Shadow Quality	High	
Smoke Generation	Low	Minimal visual benefit
Post Processing	Ultra	When using CSP, managed by PP filter

Content Manager Settings

Settings → Content Manager

- **Appearance**: Dark theme recommended for readability
- **Language**: Your preference
- **Start minimized**: Optional
- **Close to tray**: Recommended to keep CM accessible

Settings → Assetto Corsa

- **Game path**: Verify it points to your AC installation
- **Launch with CSP**: Enable after Wave 1 CSP installation
- **Audio**: Enable “Use custom audio” if you install sound mods

Force Feedback Settings

Base FFB Settings (in-game)

Setting	Recommended Value	Notes
Gain	80-100%	Adjust per car
Filter	0	No smoothing
Minimum Force	0%	Avoid artificial effects
Kerb Effects	30-50%	Personal preference
Road Effects	30-50%	Personal preference
Slip Effects	50%	Helps detect loss of grip
ABS Effects	100%	If car has ABS
Gamma	1.00	Linear response

FFB Clip Configuration

1. Launch FFB Clip before AC
2. Select your wheel from the dropdown
3. Set target clipping to 90-95%
4. Enable auto-gain adjustment
5. In-game, run 3-4 laps
6. FFB Clip will suggest gain adjustment — accept it
7. Verify: FFB bar should stay mostly green, with occasional yellow spikes

Wheel-Specific Notes

Logitech G27/G29/G920

- Logitech Gaming Software: Enable “Allow game to adjust settings”
- Overall effects strength: 100%
- Spring and damper effects: 0%
- Enable centering spring: OFF

Thrustmaster (T300/TX/T150)

- Control Panel: Set rotation to 900° (or match in-game)
- Master gain: 75-85%
- No other adjustments needed

Fanatec (CSL/DD)

- Wheel Base: SEN = Auto, FF = 100, FEI = 100
- In-game: Start with lower gain (60-70%) and increase

Content Manager Advanced Configuration

Custom Shaders Patch Settings

Accessed via CM → Settings → Custom Shaders Patch

Tab	Key Settings
General	Enable “Use 64-bit AC”, “Skip intro logos”

Tab	Key Settings
Lighting FX	Adjust sun, ambient, and contrast to preference
Weather FX	Enable “Pure weather script” after Pure installation
Particles FX	Enable smoke and rain particles
Smart Mirror	Set FPS limit to 30 to save performance

Pure Configuration

Pure is the recommended weather controller (successor to Sol). After installing Pure:

1. Launch AC through Content Manager
2. In CM → **Settings** → **Custom Shaders Patch** → **Weather FX**: Verify Pure is selected as the weather controller
3. In-game, open the Pure apps: **Pure Plan Manager** selects time and weather, **Pure Config** adjusts exposure and post-processing
4. Recommended starting preset: “3_PP_Default” for balanced visuals
5. For rain: select a rain plan and enable CSP rain effects under **CSP** → **Weather FX** → **Rain**

Audio Configuration

- **Settings** → **Audio**
- Enable “Reduce CPU load from audio” if experiencing stutters
- Set “Audio renderer” to default (Windows Audio)
- Adjust engine volume and environment volume to preference

Control Mapping

Create a profile for each control method:

1. **Settings** → **Controls** → **New Preset**
2. Name: e.g., “G29 - Default” or “Xbox Controller”
3. Map all controls precisely
4. Set steering deadzone: 0-3% (wheel) or 5-10% (gamepad)
5. Set brake/gas deadzone: 0-5%
6. Save and assign

Wheel Profiles

- **Logitech G27/G29/G920** — Overall 100%, spring/damper 0%, centering spring OFF
- **Thrustmaster T300/TX/T150** — Rotation 900°, master gain 75-85%
- **Fanatec CSL/DD** — SEN = Auto, FF = 100, FEI = 100, start gain 60-70%

Gamepad Profiles

- **Steering Speed**: 50—70% — lower values feel more stable, higher values respond faster
- **Steering Filter**: 1—5 — smooths stick input; higher = smoother but slower

- **Steering Deadzone**: 5—10% — prevents drift from worn sticks
- **Gamma**: 1.00 — linear response curve
- **Speed Sensitivity**: 0% — disables speed-dependent steering reduction
- **Vibration**: Enable for tyre slip and kerb feedback
- **Throttle Saturation**: 90—95% — leaves room for modulation at the top of the trigger range

Performance Optimization

Low-End Systems

- Reduce World Detail to Medium
- Disable mirrors (or use Smart Mirror with 15 FPS limit)
- Shadow Quality: Low
- Disable Post Processing
- Run in 1080p
- Close background applications

High-End Systems

- World Detail: Maximum
- Reflection Quality: High
- Shadow Quality: Maximum
- Post Processing: Ultra
- Enable all CSP visual features
- Use 4K resolution if supported

Appendix

Troubleshooting

Game Won't Launch

1. Verify game files in Steam (Properties → Installed Files → Verify integrity)
2. Uninstall/reinstall Content Manager
3. Delete `Documents/Assetto Corsa/cfg/` and let AC regenerate configs
4. Disable all mods in CM → launch stock AC → re-enable mods one by one

Content Manager Fails to Detect AC

1. Manually set game path in CM Settings → Assetto Corsa → Game path
2. Ensure Steam is running
3. Check that you launched AC at least once (creates registry keys CM reads)

CSP Not Working

1. Ensure CSP is installed in `assettocorsa/extensions/`
2. Check that you launched AC through CM (not Steam directly)
3. CM Settings → Assetto Corsa → enable "Use 64-bit AC"
4. Verify CSP version matches your AC version

Mods Not Appearing

1. Check the mod is installed (CM → Content tab)
2. Ensure the mod is enabled (checkbox next to it)
3. For cars: check `assettocorsa/content/cars/` for a valid folder
4. For tracks: check `assettocorsa/content/tracks/`
5. Restart CM

Performance Issues After Mod Installation

1. Lower CSP settings (especially Reflections and Shadows)
2. Disable unnecessary overlays
3. Reduce mirror resolution in CSP
4. Check for conflicting mods (see `conflicts.md`)

Force Feedback Issues

1. Verify wheel drivers are installed
2. Check that AC is selected as the game in FFB Clip

3. Reduce overall FFB gain to 70% and gradually increase
4. Set “Filter” to 0 in AC FFB settings
5. Disable “Minimum Force” and “Spring/Damper” effects in wheel software

Performance Reference

Component	Minimum	Recommended	Ideal
CPU	Intel i5-2500K / AMD FX-8150	Intel i7-4790K / AMD Ryzen 5 3600	Intel i7-10700K / AMD Ryzen 7 5800X
GPU	NVIDIA GTX 660 / AMD R9 270	NVIDIA GTX 1070 / AMD RX 580	NVIDIA RTX 3070 / AMD RX 6800
RAM	8 GB	16 GB	32 GB
Storage	HDD	SSD	NVMe SSD

Glossary

Apex — The innermost point of a racing line through a corner.

Camber — The vertical tilt of a wheel relative to the road surface. Negative camber improves cornering grip.

Content Manager (CM) — Third-party launcher and mod manager for Assetto Corsa. The primary tool for this guide.

CSP (Custom Shaders Patch) — A community-created patch for Assetto Corsa that adds advanced graphics, weather, and post-processing effects.

FFB (Force Feedback) — Haptic feedback from the steering wheel that communicates tyre grip, road surface, and car behavior.

FOV (Field of View) — The extent of the observable game world visible at any moment. Correct FOV is critical for accurate depth perception.

Oversteer — Condition where the rear wheels lose grip, causing the car to rotate more than intended.

PP Filter (Post-Processing Filter) — A visual filter applied after rendering that affects color, contrast, bloom, and tone mapping.

Pure — The current standard weather simulation system for Assetto Corsa. Replaces the older Sol system. Requires CSP 0.2.0+. Adds physically-based sky rendering, volumetric clouds, dynamic exposure, and realistic sun positioning. **Sol** — The legacy weather simulation system for Assetto Corsa that requires CSP. Adds dynamic time-of-day and realistic weather. Pure is the recommended successor.

Understeer — Condition where the front wheels lose grip, causing the car to continue straight despite steering input.

Useful Links

- OverTake.gg — Assetto Corsa Mods
- Content Manager Official Site
- Assetto Corsa Modding Guide
- r/assettocorsa on Reddit
- OverTake.gg — AC Mod Database

Migration Note: RaceDepartment → OverTake.gg

RaceDepartment was acquired by OverTake.gg. All mod download URLs now use the `overtake.gg` domain. Most mod IDs were reassigned during the migration, so links from earlier versions of this guide may need updating. If a mod URL returns a 404, search for the mod by name on OverTake.gg.

Credits

- **Author**: The Assetto Maximus team
- **Logo**: Assetto Maximus brand assets
- **Mod Credits**: Each mod's respective author(s) — see individual mod pages for full credits
- **Special Thanks**: The Assetto Corsa modding community

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